Deep Architectures Can Generalize: Sparse Feature Selection Improves Spatial Transfer in Earth Observation

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Declarations

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Abstract

Accurate crop classification from satellite imagery is essential for agricultural monitoring, yield forecasting, and food security assessment, yet remains particularly challenging in dryland farming regions of the Global South where spectral similarity between crop types, irregular phenological cycles, and persistent cloud cover complicate classification. In this critical context we test and propose a number of experiments relevant to agriculture in the Global South under the conditions of data and methodological constraints. Most crop-classification studies report performance under random or field-wise cross-validation *within* a single region, which measures in-distribution fit rather than the spatial transfer that operational mapping actually requires. Using a labeled field-boundary dataset from the Western Cape of South Africa, we benchmark twenty-three classical, deep learning, and patch-based models twice: under conventional in-region field-wise validation, and on a spatially disjoint holdout tile. The two protocols produce markedly different conclusions. Under in-region validation the dense temporal deep nets cluster at the top (macro-F1 0.72–0.78); under spatial transfer this cluster fractures: the two in-region leaders (L-TAE and a Transformer encoder, both 0.78 at the pixel level) drop to 0.58, and CNN-BiLSTM, singled out as best by prior analysis of this dataset, suffers the largest degradation of any model (macro-F1 0.72 to 0.46, a 0.26 drop), landing among the worst absolute scores. The ranking inverts: a TabNet ensemble becomes the best single model (macro-F1 0.60, Kappa 0.54), and simple classical baselines such as gradient-boosted trees and even logistic regression (macro-F1 0.56) are among the most robust transferers. We argue this is not incidental but mechanistic: the sparse, axis-aligned feature selection that made Random Forest and gradient-boosted trees the remote-sensing workhorse for two decades is precisely the inductive bias that governs spatial transfer, and TabNet inherits it (attentive sparsemax feature masks) while dense CNN/RNN/attention temporal nets discard it and overfit region-specific signal. Controlled architectural comparisons isolate the cause: a plain Transformer encoder, the most expressive attention model we test, transfers no better than the lightweight L-TAE, so attention capacity is not the lever, whereas adding a sparse feature-selection gate is. We further show this inductive bias can be grafted onto a deep network: a sparse-attention encoder we introduce (L-TAE-S), which adds TabNet-style sparsemax feature masks to the L-TAE, ties TabNet for the best transfer and posts the smallest in-region-to-holdout gap of any model. A complementary lever, field-level aggregation as variance reduction, recovers transfer for dense models (L-TAE improves when aggregated) and composes with the sparse gate in L-TAE-S to yield the smallest transfer gap, but hurts TabNet, whose sparse gate exploits the per-pixel observation volume that aggregation removes. We further show, via a training-set reduction experiment, that the most transferable models retain near-full holdout accuracy with as little as 25% of the training fields. We also compare rich feature sets with classical models, including XGBoost, Random Forest, and LightGBM, via automated time-series feature set created with `xr_fresh`, evaluated here for the first time in crop classification. The central message for the remote-sensing community is twofold: model selection in crop mapping is an artifact of the validation

protocol (spatial transfer, not in-distribution accuracy, must drive it), and the property that decides whether a model survives that transfer is sparse, axis-aligned feature selection.

Keywords: Crop classification, Satellite imagery, Deep learning, Machine learning, Remote sensing

1 Introduction

Accurate and timely crop classification is foundational to precision agriculture, enabling yield forecasting, resource optimization, food security assessment, and climate-resilient decision-making [54, 1]. The widespread availability of high-resolution Earth observation data, especially from the Sentinel-2 satellite constellation, has transformed the landscape of agricultural monitoring. With 13 spectral bands, spatial resolutions ranging from 10 to 60 meters, and a revisit frequency of approximately five days, Sentinel-2 offers dense multi-temporal imagery rich in spatial and spectral information.

Despite these advancements, reliable crop classification remains a complex task on plots throughout Africa and the Global South. Spectral similarities between crops, temporal variability in planting and phenological cycles, and atmospheric disturbances such as cloud cover hinder classification accuracy. In Africa, this is compounded by the complexities of rain-fed farming [54]. Traditional machine learning methods often rely heavily on manually engineered features, such as vegetation indices or statistical summaries, which may capture pertinent temporal and spectral dynamics but are often more limited in complexity of features. A number of methods like XGBoost (XGB) and Random Forest (RF) have proven effective across crop classification and broader agricultural prediction tasks such as yield and loss estimation [41, 10, 42, 43]. However in regions with significant differences in cropping patterns and practices, the complexity of the task can outstrip the exploitable variability in the training data [44, 41]. Newly released automated feature extraction methods like that implemented in `xr_fresh` offer the possibility of rich feature sets for use in machine learning but it remains untested in the literature [44]. Despite their effectiveness, machine learning methods can be inconsistent [30]. Several factors contribute to this variability, including hyperparameter selection [11], sample quality and quantity [12], selection of training and testing splits [14], imbalanced classes [14], and feature generation and selection [13].

To address these limitations, deep learning approaches have gained traction in crop type mapping, with both Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) emerging as promising architectures. RNNs are well suited to the sequential nature of remote sensing time series, as their internal hidden states allow them to retain a form of memory across time steps, capturing the temporal dynamics of crop phenology that static or summary-based features often miss. This capacity to model temporal structure has proven especially valuable for crop classification in data-rich environments, where the full growing season trajectory of spectral signals can be exploited [45, 31]. The lack of high quality training data in the Global South and lack of reliable cloud-free imagery has pushed many of the most successful efforts into the realm of precision agriculture such as applications for hyperspectral and drone-derived imagery [41]. Nevertheless, recent work applying deep learning to agricultural mapping in developing world contexts has demonstrated encouraging results. Ayushi et al. [47] applied a U-Net fully convolutional encoder-decoder architecture to a crop type mapping task in Kenya, achieving a macro-F1 of roughly 0.74, while Nowakowski et al. [53] fine-tuned pre-trained convolutional networks via transfer learning for crop-type mapping in Malawi, reaching around 83% accuracy. Rustowicz et al. [38] released the first crop-type semantic-segmentation dataset for smallholder farms in Ghana and South Sudan, underscoring the difficulty of small fields, frequent cloud cover, and scarce labels in such settings. However questions remain about the generalizability of deep learning models particularly in complex environments with limited training data [49].

Hybrid approaches offer a compelling opportunity by combining the representational power of deep learning with the interpretability and efficiency of traditional machine learning, and several studies have demonstrated their value in agricultural remote sensing contexts. A common strategy in precision agriculture involves using a pre-trained deep learning model as a feature extractor especially, with the resulting high-level representations passed to a traditional classifier for the final prediction. Espejo-Garcia et al. [50] fine-tuned architectures including MobileNet, DenseNet, and Xception before replacing their fully connected layers with shallow classifiers such as Support Vector Machines, gradient boosting, and logistic regression for weed identification, while Koklu et al. [51] found that, on a single grapevine-leaf dataset, pairing MobileNetV2 feature extraction with an SVM improved on end-to-end deep learning. Yang et al. [52] more relevant to this study, combined a CNN with a Random Forest classifier for Sentinel-2 crop type mapping, leveraging the spatial-spectral learning capacity of the CNN alongside the robustness and resistance to overfitting characteristic of ensemble tree methods. Taken together, these hybrid approaches suggest a productive middle path between the interpretability and data efficiency of classical machine learning and the feature-learning capacity of deep neural networks, and they form part of the methodological landscape against which the approach taken in this study is positioned.

This study addresses a critical yet underexplored gap in the crop-classification literature: how remote-sensing models behave not under the data-rich, temperate conditions where most methods are developed and validated, but under the conditions that actually govern operational agricultural monitoring across the Global South. Crop mapping in these regions underpins food-security assessment, yield forecasting, and climate-resilience planning for billions of people, yet it must contend with scarce and unevenly distributed crop labels, heterogeneous and spatially variable smallholder and traditional management, irregular planting calendars, and persistent cloud cover. The Western Cape of South Africa concentrates these challenges, including smaller field sizes, rain-fed dryland farming systems, heterogeneous and spatially variable planting rotations, persistent cloud cover during the critical winter growing season, and phenological cycles that do not conform to the assumptions embedded in many widely used models, while offering the rare advantage of an unusually rich, high-quality labeled dataset. This combination makes it an unusually realistic test bed: rich enough to train and compare modern architectures fairly, yet hard enough to expose the failure modes that matter in practice. Crucially, operational mapping in such settings means applying a model to regions where no labels were collected, so the realistic measure of success is not in-region accuracy but spatial transfer, precisely the evaluation that most of the literature omits. We leverage this combination to ask not merely which method scores highest, but which methods transfer to unseen terrain, why, and how little labeled data they require. Our experiments utilize a labeled field-boundary dataset provided by Radiant Earth's MLHub [2], capturing ground-truth crop types for agricultural fields in South Africa during the 2017 growing season.

We evaluate multiple deep learning architectures, including CNN-BiLSTM hybrids, TabNet ensembles, light-weight temporal-attention (L-TAE) and full Transformer-encoder networks, temporal-convolution (TempCNN) networks, and 3D CNN models, that extract dynamic phenological features across time, and benchmark these against traditional ensemble methods including XGBoost, LightGBM, and Random Forest. Critically, we evaluate every model twice: under the conventional protocol, random or field-wise cross-validation *within* the training region, and on a spatially disjoint holdout tile that no training pixel touches. This dual evaluation exposes a large and architecture-dependent spatial-transfer gap that single-region validation hides entirely, and it reorders the models. Throughout, we restrict our inputs to multi-temporal Sentinel-2 optical imagery alone, without recourse to SAR data, in order to isolate the signal available from spectral time series and maximize applicability to contexts where only optical data is available or technical capacity is limited. All code is publicly available to support reproducibility and future research [4].

This study makes five contributions. First, a multi-architecture benchmark (twenty-three classical, deep, and patch-based models) showing that in-region versus spatial-transfer evaluation produce *inverted* rankings on the same data. Second, a mechanistic account of *why*: sparse, axis-aligned feature selection, the property behind two decades of Random Forest and gradient-boosted-tree dominance, is what survives the move to a holdout tile, and TabNet inherits it via sparsemax masks. Third, a controlled architectural comparison isolating sparse feature selection rather than attention capacity as the lever, including L-TAE-S, a sparse-attention encoder we introduce that ties TabNet for the smallest transfer gap of any model. Fourth, an identification of field-level aggregation as a second, substitutable robustness lever that recovers transfer for dense models but is redundant for sparse ones. Fifth, a training-set reduction experiment and the first evaluation of `xr_fresh` automated time-series feature extraction in crop classification. Taken together, these make the work at once a benchmark, a methodological, and a spatial-transfer study: we benchmark twenty-three models under two evaluation protocols, and use controlled architectural variants to identify *why* transfer succeeds or fails.

This leads to the central research question: across a range of machine learning, deep learning, and hybrid architectures, which approach most reliably classifies crop types from multi-temporal Sentinel-2 optical imagery in a complex dryland farming environment *when evaluated on a spatially disjoint region*, and what properties of a model determine whether its in-distribution accuracy survives the transfer to unseen terrain?

1.1 Related Work

Crop classification using satellite imagery has become a core task in precision agriculture because it supports large-scale crop monitoring, yield estimation, resource management, and food-security analysis. The increasing availability of high-resolution, multi-temporal Earth observation data, particularly from Sentinel-2, has enabled substantial progress in crop mapping, but several challenges remain. These include strong spectral similarity among crop types, temporal variability in planting and phenological cycles, class imbalance, and the difficulty of building methods that generalize across regions and farming systems.

Early work in crop classification relied primarily on classical machine learning methods such as Random Forest, Support Vector Machines, Logistic Regression, and gradient-boosted tree models. These approaches typically operate on hand-crafted inputs derived from spectral bands, vegetation indices, temporal summaries, or other engineered descriptors, and they have often proven effective because they are computationally efficient, interpretable, and robust on structured remote-sensing data [30, 14, 26, 28, 27]. However, their performance depends heavily on feature engineering and domain knowledge, which can limit transferability across crop systems and agroecological regions. This challenge becomes especially pronounced in heterogeneous agricultural landscapes, where differences in management practices, field conditions, and phenological timing can reduce the separability captured by manually designed features [30, 24, 41]. Comparative studies have directly weighed classical machine learning against deep learning for crop identification, albeit largely on close-range crop imagery rather than satellite scenes [15].

These limitations have motivated growing interest in deep learning approaches that learn hierarchical feature representations directly from raw or minimally processed observations. Convolutional neural networks (CNNs) are effective at extracting local spectral and spatial patterns [32], while recurrent architectures such as LSTMs and BiLSTMs are well suited to modeling temporal crop-development signals over a season [1, 17, 9]. More broadly, deep learning has become increasingly important in remote sensing because it reduces reliance on manually constructed predictors and can capture complex temporal structure that summary statistics often miss [31, 37]. Prior work has shown that temporal modeling is especially valuable when crop types are difficult to distinguish from single-date imagery but follow different seasonal trajectories [17, 24].

A particularly important direction in the literature has been the development of hybrid architectures that combine convolutional feature extraction with temporal sequence modeling. For example, Fan et al. proposed a hierarchical convolutional recurrent neural network for Sentinel-2 image classification, showing that combining CNN-based feature extraction with recurrent modeling can improve in-region classification accuracy relative to both classical and standalone deep learning baselines [7]. Similarly, Tufail et al. demonstrated that hybrid CNN-RNN models applied to multi-temporal Sentinel-2 imagery and red-edge vegetation indices can achieve very strong crop-mapping performance by jointly learning spatial context and temporal phenology [8]. Bandar and Coşkunçay likewise showed that combining an attention-based

BiLSTM with a temporal CNN improves crop classification from Sentinel-2 time-series data, reinforcing the broader value of spatiotemporal representation learning in agricultural remote sensing [9].

Beyond fully end-to-end deep architectures, several studies have explored hybrid pipelines that combine learned representations with classical machine learning classifiers. Espejo-Garcia et al. used deep convolutional backbones as feature extractors before applying conventional classifiers for weed identification, while Koklu et al. found that deep features combined with an SVM improved on end-to-end models on a grapevine-leaf dataset [50, 51]. In a setting more directly related to this study, Yang et al. combined CNN-based representation learning with a Random Forest classifier for multi-temporal crop classification from Sentinel-2 imagery [52]. Taken together, these studies suggest that hybrid approaches can provide a useful middle ground between the flexibility of deep learning and the robustness and interpretability of classical models.

The representation of input data is another central theme in the literature. Sentinel-2 has become a preferred platform for crop monitoring because of its high revisit rate and multispectral richness, and these data are exploited either through engineered descriptors (spectral indices, temporal summaries) fed to classical learners or through end-to-end models that learn directly from temporal reflectance sequences. A recurring assumption is that richer, learned representations should generalize better than compact engineered ones; whether this holds under genuine spatial transfer (as opposed to within-region validation) is less well established and is a central question of this study.

A second, and for our purposes critical, gap concerns *how generalization is measured*. Much of the crop-classification literature reports accuracy under random or field-wise cross-validation drawn from a single region or scene [14, 35, 37, 17, 1]. Such protocols hold out individual pixels or fields but not *space*: training and test observations share the same atmospheric conditions, soils, management calendars, the farmers themselves, and sensor-acquisition geometry, so the reported numbers reflect in-distribution fit rather than transfer to new terrain. Because operational crop maps are produced precisely by applying a model to regions where no labels were collected, this distinction matters, yet studies that re-rank models under a spatially disjoint hold-out remain rare [49, 12, 38]. A handful of architectures explicitly target cross-scenario generalization: Cropformer, for example, couples convolutional and transformer features with self-supervised pre-training to handle full-season, in-season, few-sample, and transfer-learning settings, including a cross-region evaluation [39], though such cross-scenario architectures remain the exception, and are rarely benchmarked against classical and deep baselines on a common spatially disjoint holdout. This is compounded in the Global South, where rain-fed farming, heterogeneous field conditions, irregular planting calendars, and persistent cloud cover make spatial transfer especially demanding [54].

Our study is positioned within this gap. Rather than evaluating a single architecture in isolation, we conduct a systematic comparison of classical machine learning, deep learning, and patch-based approaches under *both* in-region field-wise validation and a spatially disjoint holdout, in the Western Cape of South Africa. We build on prior work in temporal modeling, hybrid learning, and ensemble-based classification, while also evaluating `xr_fresh` as an automated time-series feature-extraction framework in a crop-classification setting [44]. This allows us to ask not only which approach is most accurate, but which model properties cause in-distribution accuracy to survive, or fail to survive, the move to unseen terrain.

2 Data Description

The dataset used for this study consists of two primary components: satellite imagery and field boundary data, as sourced from [22].

2.1 Study Area

The study was conducted in an agriculturally productive region of the Western Cape of South Africa, characterized by diverse agroecological conditions and cropping systems. Major crop types in the dataset include wheat, barley, canola, small-grain grazing, and lucerne/medics (alfalfa). The region spans varying climatic zones, from semi-arid to subtropical, and exhibits seasonal differences in rainfall and temperature, contributing to distinct crop phenologies.

The labeled data are organized into Sentinel-2 UTM grid tiles (zone 34S). Two adjacent tiles (`34S_19E_258N` and `34S_19E_259N`) form the training region, comprising 4,150 labeled fields covering 622 km^2 of cultivated area (mean field size 15.0 ha, median 11.0 ha). A third tile (`34S_20E_259N`), immediately east of and adjacent to the training tiles, is reserved as a holdout region for spatial-transfer evaluation and contains 2,417 labeled fields over 282 km^2 (mean 11.7 ha, median 6.9 ha). The tiles do not overlap, so no pixel or field is shared between training and holdout, and transfer to the holdout must contend with different soils, field geometries, and acquisition conditions. Because the holdout nonetheless lies within the same agroecological zone and growing-season calendar as the training tiles, this is a relatively *local* spatial shift; the spatial-transfer gaps we report should accordingly be read as a conservative lower bound on the degradation expected under transfer across regions or climates (Section 5). Unless otherwise noted, "in-region" results refer to a field-wise train/validation/test split within the training region, while "spatial-transfer" or "holdout" results refer to predictions on the disjoint tile.

Field parcels differ in size, shape, and management practices, introducing heterogeneity into the dataset. Sentinel-2 imagery was collected monthly across the 2017 growing season, covering key phenological stages from emergence to harvest. Each field carries a single crop-type label for the season. This temporal resolution is vital for capturing crop-specific growth patterns, which are often indistinguishable in single-date imagery.

2.2 Remote Sensing Data

Sentinel-2 Level-2A optical surface-reflectance data were acquired over the study area in South Africa for the January–December 2017 growing season. Scenes were first filtered to a reported cloud cover of $\leq 30\%$ using the ESA `CLOUDY_PIXEL_PERCENTAGE` metadata, and then per-pixel cloud-masked using the `s2cloudless` cloud-probability product at an 80% probability threshold, combined with a morphological cloud-shadow mask derived from NIR-darkness projection and dilated by a 90 m buffer. Six features were produced for each month: four optical bands, namely B2 (Blue, 490 nm), B6 (Red Edge 2, 740 nm), B11 (SWIR 1, 1610 nm), and B12 (SWIR 2, 2202 nm), together with the Enhanced Vegetation Index (EVI) and Hue, a colorimetric descriptor of vegetation condition [10]. The reflectance bands were formed as monthly median composites, whereas EVI and Hue were computed *per scene* and then composited to a monthly value (detailed below). All layers were scaled to integer reflectance ($\times$10,000) and exported as 10 m GeoTIFFs via Google Earth Engine. This band subset was chosen to span the spectral regions most informative for separating the target crops while limiting redundancy: the blue band (B2) constrains atmospheric and soil-brightness effects and enters the EVI formulation; the red-edge band (B6) is sensitive to chlorophyll content and canopy structure, which differ across cereals and broadleaf crops; and the two SWIR bands (B11, B12) respond to leaf and soil moisture and to senescence dynamics that help discriminate phenologically similar cereals such as wheat and barley. Red-edge bands in particular are repeatedly reported among the most informative Sentinel-2 features for separating spectrally similar crops [8, 41]. We deliberately did not retain the red (B4) and near-infrared (B8) bands as standalone features. Although these are the classical vegetation bands, their information is not discarded: it enters the feature set through EVI (which is itself a function of near-infrared, red, and blue reflectance) and through the chlorophyll-sensitive red-edge band (B6), which is spectrally adjacent to the near-infrared. Prior feature-importance analysis on comparable Sentinel-2 crop data motivated this compact subset as a way to limit redundancy among highly correlated bands [10]. Importantly, EVI and Hue were computed from the complete per-scene reflectance before this subset was retained, so the index values are unaffected by the band selection.

Eleven monthly composites were produced over the growing season; June 2017 yielded no usable imagery and May 2017 was discarded due to persistent cloud cover, leaving 10 valid monthly observations per feature. Each pixel was consequently represented by a 60-dimensional time-series feature vector (six features across ten months), enabling the capture of rich phenological dynamics across the season.

A key methodological point concerns the order of operations for the derived indices. EVI and Hue were computed *per scene* (that is, from the simultaneously acquired bands of each individual Sentinel-2 acquisition) and only then composited, by taking the monthly median over the cloud-free per-scene index values. This is distinct from, and avoids the pitfall of, computing an index from already-composited bands: because each per-scene EVI uses bands observed at the same instant, the per-acquisition band simultaneity that the EVI formulation assumes is preserved, and the resulting monthly EVI is a physically consistent index rather than an artifact of mixing reflectances drawn from different dates. The monthly median then serves only to reduce residual cloud and acquisition noise across the index time series, exactly as it does for the raw reflectance bands.

By relying exclusively on optical data rather than fusing it with SAR imagery [6, 48], this study demonstrates the capability of multi-temporal spectral analysis alone for robust crop classification. This approach is directly transferable to emerging high-resolution optical platforms such as PlanetScope, reinforcing the operational relevance of dense, temporally-aware optical remote sensing for agricultural monitoring at scale.

2.3 Field Boundary Data

The field boundary polygons were drawn from the crop type classification dataset for Western Cape, South Africa [2]. This GeoJSON file delineated each agricultural field as a polygon, with boundaries digitized primarily from 50 cm aerial photography at scales of 1:2,500 for horticultural parcels and 1:7,500 for other fields between May and August 2017. In cases where ground reference was unavailable, Sentinel-2 imagery supplemented the digitizing process. Each polygon fully encompassed the field area and carried a crop-type label obtained via aerial and vehicle surveys conducted from May 2017 to March 2018 [2]. All predictions in our study were made at the field level to reflect the targeted application.

2.4 Crop Types and Distribution

In this study, we classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley, and small-grain grazing. Figure 1 illustrates the distribution of crops in the study area. The data are notably imbalanced: in the training region lucerne/medics accounts for 1,792 of the 4,150 fields (43%), followed by wheat (753), barley (660), canola (512), and small-grain grazing (433); the holdout region is more skewed still, with lucerne/medics comprising 1,360 of 2,417 fields (56%). This imbalance has important implications for model training and evaluation, as classification algorithms may be biased towards the majority class unless appropriate balancing techniques are applied, and it motivates our use of macro-averaged F1 as the primary metric (Section 3.3). The stakes of this choice are visible in independent transfer studies: under cross-domain transfer Mai et al. [40] report a weighted-F1 of 0.76 alongside a macro-F1 of just 0.45, the gap reflecting minority crops largely absorbed into the majority class, so a weighted or overall-accuracy summary would have declared "successful" a transfer that macro-F1 reveals to be poor.

Figure 2 presents the mean field area for each crop type, measured in hectares. Although lucerne/medics represent the majority of fields, their average area (11.6 ha) is the smallest of the five crop types.

Figure 3 displays box plots of spectral band intensities for each crop type, illustrating the distribution and overlap of spectral responses.

Although some differences in median and spread are evident, the substantial overlap in spectral intensities across crops demonstrates that classification based solely on a limited set of spectral bands is challenging and prone to confusion. This was particularly true for cereal crops such as wheat and barley, which exhibited highly similar spectral profiles, as shown in Figure 4. The close alignment of their spectral signatures underscores why models often struggle to distinguish between these two crops using spectral data alone [23].

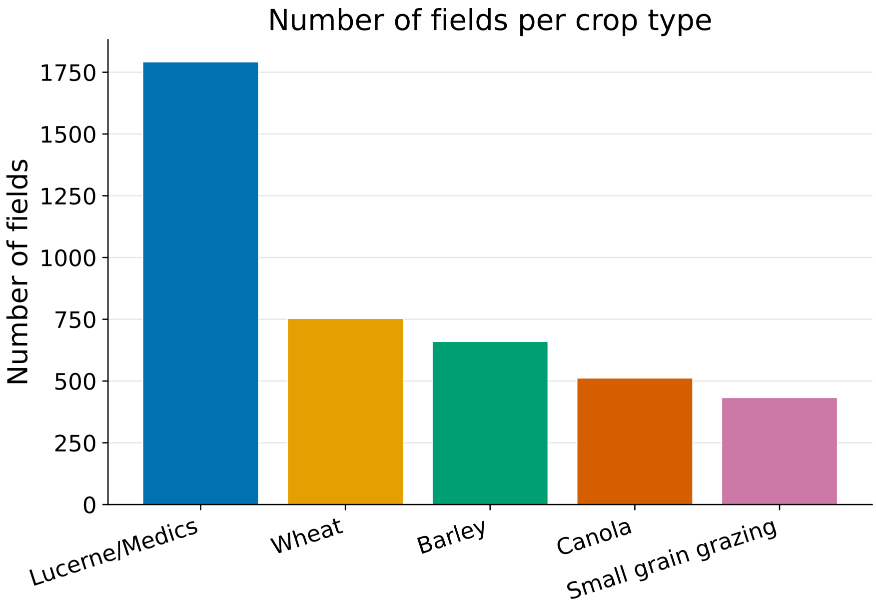

Fig. 1: Number of fields per crop type.

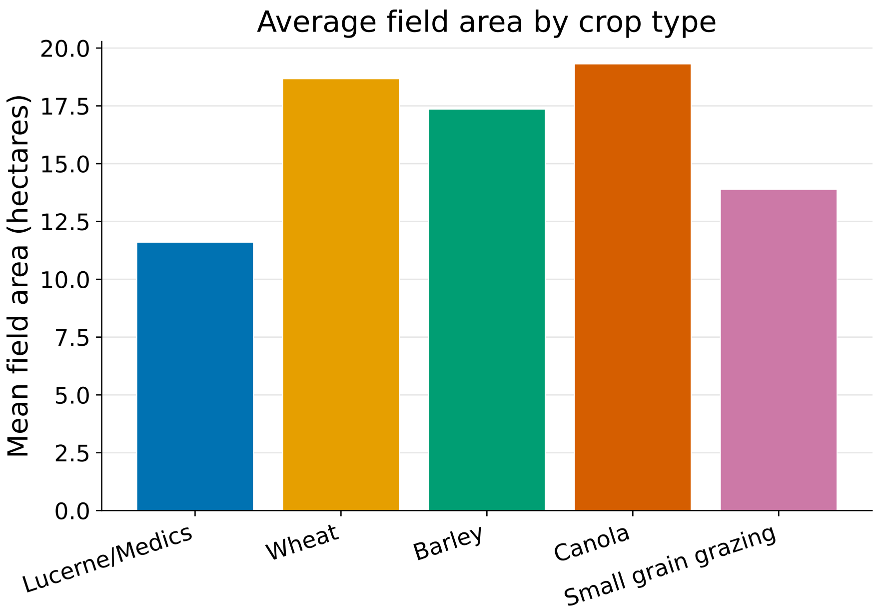

Fig. 2: Average field area by crop type (hectares).

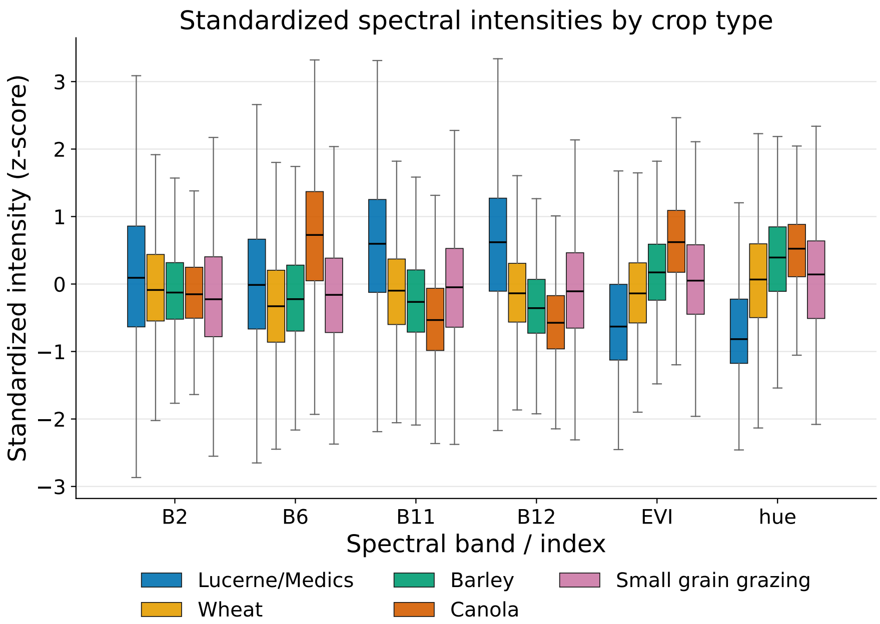

Fig. 3: Spectral band intensities by crop type.

Boxplots show the distribution and overlap of spectral responses across crops.

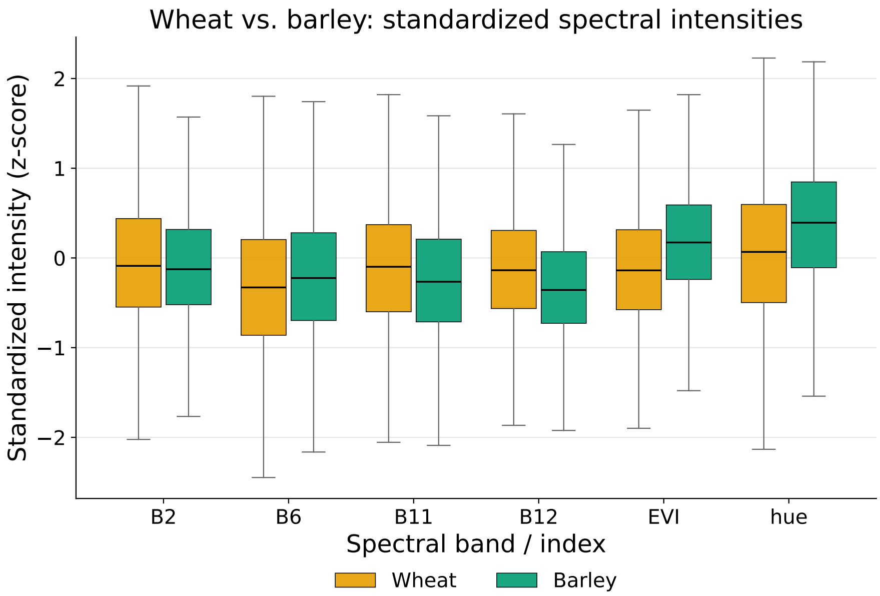

Fig. 4: Wheat versus barley spectral band intensities.

The two cereals show a high degree of spectral similarity, motivating their persistent classification confusion.

These findings highlight the need to incorporate temporal information that better represents phenological stages throughout the growing season to improve crop classification. The unique temporal dynamics associated with the stages of crop development (e.g., emergence, peak greenness, senescence) provide additional discriminatory power that can help resolve ambiguities between crops with similar spectral signatures [24]. Therefore, integrating both spectral and temporal data was crucial to enhance classification accuracy, especially in complex agricultural landscapes where crops like wheat and barley were spectrally similar.

3 Methodology

This study explores three complementary modeling paradigms for crop classification: feature-based machine learning, deep learning, and patch-based approaches. These paradigms differ in how spatial and temporal information from multi-spectral Sentinel-2 imagery is represented and utilized.

Classical machine learning relies on engineered features derived from pixel-level time series, while deep learning models learn hierarchical representations directly from raw data. In contrast, patch-based approaches operate on spatially contiguous regions, enabling the model to capture local spatial structure and contextual relationships that are not available in purely pixel-wise representations. Each paradigm was evaluated at both the pixel level and the field (or patch) level to assess their effectiveness across different representations of the data. Crop classification from remote sensing data can therefore be approached through three complementary strategies: feature engineering, representation learning, and spatial context modeling.

In the feature engineering paradigm, domain knowledge is used to construct descriptive inputs from raw spectral time series. These include vegetation indices such as NDVI, spectral band ratios, textural measures, and phenological statistics, which are then used as inputs to classical machine learning models such as Random Forests (RF), Support Vector Machines (SVM), and Gradient Boosted Trees (GBT) [1]. In this study, we adopt the `xr_fresh` framework to automatically extract a comprehensive set of time-series features, enabling the construction of high-dimensional representations without manual feature design [44].

In contrast, deep learning follows a representation learning paradigm, where models learn hierarchical feature representations directly from raw input data. Convolutional Neural Networks (CNNs) capture local spatial and spectral patterns, while Recurrent Neural Networks (RNNs) [16] and Transformer-based architectures [39] model temporal dependencies and long-range interactions in multi-temporal data. These approaches reduce reliance on hand-crafted features and allow the model to learn complex crop growth dynamics directly from the data.

Beyond pixel-level modeling, patch-based approaches introduce an additional spatial dimension by processing contiguous regions of imagery rather than individual pixels. This enables models to capture local texture, spatial heterogeneity, and contextual relationships between neighboring pixels, which are particularly important in agricultural landscapes where crop patterns exhibit strong spatial coherence. Patch-based models therefore complement both classical and deep learning approaches by explicitly incorporating spatial context into the classification process.

This study systematically compares these paradigms to evaluate their relative strengths, computational trade-offs (per-model training times are tabulated in Appendix C), and effectiveness in handling complex agricultural environments, particularly under conditions of spectral similarity and temporal variability. Model performance was assessed using a combination of standard classification metrics: macro-averaged F1 (our primary metric), Cohen's Kappa, weighted F1, and log-loss (cross-entropy). These metrics were selected to evaluate not only overall accuracy but also class balance across imbalanced crop types, and to express out-of-sample performance on the source challenge's official field-level cross-entropy metric; they are motivated in Section 3.3 and formally defined in Appendix B.

3.1 Classical Machine Learning

In the classical approach, we automated feature extraction using the `xr_fresh` toolkit to rapidly compute a comprehensive set of statistical and temporal features from each 10-month spectral time series (Figure 5) [10, 44]. To address class imbalance, we applied SMOTE. Several classifiers such as Logistic Regression, Random Forest, LightGBM, and XGBoost were then evaluated.

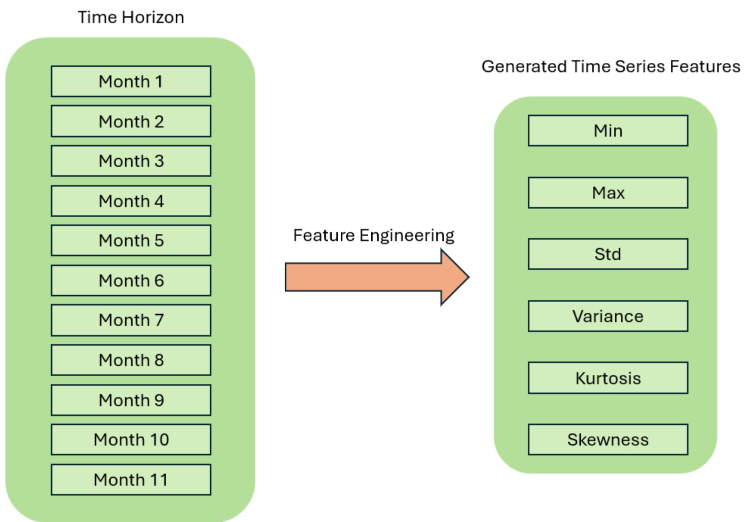

Fig. 5: Extraction of statistical and temporal features from the 10-month spectral time series.

Table 1 lists the time-series features extracted from each pixel's 10-month spectral profile [10]. This toolkit allowed us to extract all required time-series features efficiently, ensuring each pixel's temporal signature was well-represented for robust classification [44].

Table 1: Extracted time-series features.

Feature	**Description**
Absolute Energy	Sum of squared values, capturing overall signal magnitude.
Absolute Sum of Changes	Total absolute difference between consecutive measurements, reflecting temporal variability.
Autocorrelation (1, 2-month lag)	Correlation with its one- and two-month lagged versions, indicating series persistence.
Count Above Mean	Number of points above the series mean, quantifying high-value occurrences.
Day of Year of Max/Min	Julian day when the maximum and minimum occur, marking peak and low activity periods.
Kurtosis	Tail heaviness of the distribution, highlighting propensity for extremes.
Linear Trend	Slope of a least-squares fit, summarizing the overall upward or downward trend.
Longest Strike Above/Below Mean	Longest consecutive run above or below the mean, capturing sustained periods.
Max/Min	Absolute extreme values in the series.
Mean/Median	Measures of central tendency.
Mean Absolute Change/Mean Change	Average magnitude and signed change between consecutive points.
Mean Second Derivative	Mean of the series' second differences, detecting acceleration or deceleration.
Quantiles (0.05, 0.95)	Values at the 5th and 95th percentiles, summarizing distribution extremes.
Ratio Beyond $r\sigma$	Proportion of points more than r standard deviations ($r = 1, 2, 3$) from the mean.
Skewness	Asymmetry of the distribution around the mean.
Standard Deviation/Variance	Spread measures around the mean.
Sum of Values	Total sum of observations (mean $\times$ length).
Symmetry Looking	Similarity when the series is reversed.
Time Series Complexity	Complexity-Invariant Distance metric of sequence irregularity.

Time-series features extracted from each pixel's spectral profile with `xr_fresh`.

3.1.1 Pixel-Level Analysis

For pixel-level analysis, each pixel was treated as an independent observation. Feature vectors from each pixel's 10-month history were used to train classifiers. During inference, pixel predictions within a field polygon were aggregated via majority voting to determine the overall field label.

Baseline Classical Models.

To benchmark crop classification, we used four classical models: Logistic Regression, Random Forest, LightGBM, and XGBoost, ranging from interpretable linear models to advanced tree-based learners. Each pixel was represented by a feature vector $\mathbf{x}_i \in \mathbb{R}^{174}$, formed from time-series features across ten months and six spectral bands and indices (B2, B6, B11, B12, EVI, Hue), capturing phenological signatures over time. These four models are standard and we summarize only their configurations here; their formal definitions are collected in Appendix A. Logistic Regression [25] is a linear multinomial classifier providing interpretable per-class weights. Random Forest [26] is a bagged ensemble of decision trees whose axis-aligned splits capture non-linear interactions; we used 300 trees. LightGBM [27] and XGBoost [28, 29] are gradient-boosted tree ensembles with leaf-wise growth and explicit L_1/L_2 regularization; both were trained with GPU histogram methods. All four operate on the same per-pixel `xr_fresh` feature vector and, being tree- or margin-based with built-in feature selection or shrinkage, serve as the parsimonious baselines against which the deep models are compared.

Pixel-Level Ensemble with Hybrid Voting.

We ensembled RF, XGBoost, and LightGBM using soft voting. Per-pixel class probabilities were averaged:

$$\hat{\mathbf{P}}_i = \frac{1}{3}\left(\mathbf{P}_{\mathrm{RF}}(\mathbf{x}_i) + \mathbf{P}_{\mathrm{XGB}}(\mathbf{x}_i) + \mathbf{P}_{\mathrm{LGBM}}(\mathbf{x}_i)\right) \quad \hat{y}_i = \arg\max_k \hat{P}_i(k)$$

Field-Level Aggregation.

Let field f consist of pixels $i \in f$. Field-level soft voting is computed as:

$$\bar{\mathbf{P}}_f = \frac{1}{|f|}\sum_{i \in f} \hat{\mathbf{P}}_i \quad \hat{y}_f^{\mathrm{soft}} = \arg\max_k \bar{P}_f(k)$$

We introduced a confidence margin:

$$\Delta_f = \bar{P}_f^{(1)} - \bar{P}_f^{(2)}$$

where $\bar{P}_f^{(1)}$ and $\bar{P}_f^{(2)}$ are the highest and second-highest class probabilities in $\bar{P}_f$, respectively.

If $\Delta_f < \theta$ (threshold $= 0.10$), we fall back to mode voting across pixel labels:

$$\hat{y}_f^{\mathrm{mode}} = \mathrm{mode}\,\{\hat{y}_i \mid i \in f\}$$

Final Hybrid Rule.

$$\hat{y}_f = \begin{cases} \hat{y}_f^{\text{soft}}, & \text{if } \Delta_f \geq \theta \\ \hat{y}_f^{\text{mode}}, & \text{otherwise} \end{cases}$$

This strategy balanced probabilistic confidence with categorical consistency, improving field-level classification stability.

3.1.2 Field-Level Analysis

In the field-level analysis, we aggregated all pixel features within each field polygon, as shown in Figure 6, by computing the mean across pixels to form a single representative feature vector per field. This step reduced noise and the influence of outlier pixels while preserving meaningful spatial patterns. Classifiers were then trained using these aggregated field-level vectors to directly learn from spatially coherent crop representations. This approach aligns with previous findings that object-based aggregation helps in minimizing spectral confusion and enhances classification accuracy, especially in homogeneous agricultural regions [13].

To address class imbalance at the field level, we combined SMOTETomek resampling with a stacked ensemble classifier. This approach ensured both balanced data and strong generalization performance.

SMOTETomek Resampling.

SMOTETomek combines oversampling and undersampling techniques. SMOTE generates synthetic samples for underrepresented classes by interpolating between existing samples, while Tomek Links removes borderline examples to improve class separability. This helped mitigate the severe class imbalance visible in Figure 1, especially for crops like small-grain grazing and canola.

Stacked Ensemble Classifier.

We built a two-layer ensemble consisting of Random Forest and XGBoost as base learners, with Logistic Regression as the meta-learner, allowing it to use both base-model predictions and the original input features. The final prediction for pixel i is defined as:

$$\hat{y}_i^{\text{stacked}} = f_{\text{meta}}\left(f_{\text{RF}}(\mathbf{x}_i),\ f_{\text{XGB}}(\mathbf{x}_i),\ \mathbf{x}_i\right)$$

This architecture combines the non-linear modeling capacity of tree-based learners with a linear meta-learner that integrates both base predictions and original input features.

We report this stacked ensemble in two forms that share the architecture above and differ only in resampling: *SMOTE Stacked*, trained on SMOTETomek-resampled data as described, and *Stacking*, the identical pipeline without resampling, so the pair isolates the effect of class rebalancing on spatial transfer.

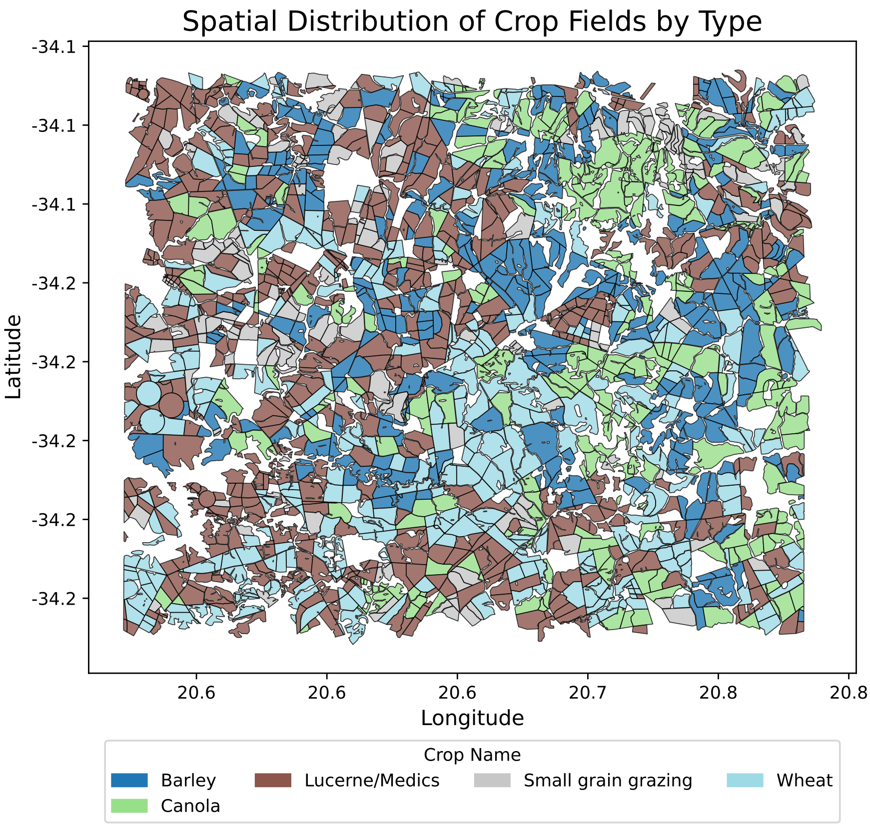

Fig. 6: Field-level aggregation of pixel features.

Pixel values are grouped by field polygon and summarized into a single feature vector per field.

Field-Based Aggregation and XGBoost Optimization.

We also implemented an optimized field-level pipeline using XGBoost. Field identifiers were used to ensure that no field contributed to more than one data split, with fields split into train/validation/test sets at the field level. Features were aggregated via the mean across pixels, and labels via the mode crop type per field.

Optuna was then used to tune XGBoost over 150 trials, searching over the number of trees, tree depth, learning rate, row and column subsampling fractions, minimum split loss, minimum child weight, and L_1 and L_2 regularization strengths, with the objective of maximizing the macro-averaged F1 score via 5-fold stratified cross-validation. The best configuration (Trial 32) is detailed in Table 2, which also achieved a validation accuracy of 76.61%.

3.2 Deep Learning Architectures

Deep learning models ingest raw spectral values and learn hierarchical representations that capture both spatial and temporal dependencies, offering an alternative to the

Table 2: XGBoost hyperparameter search results.

Trial	Acc.	Est.	Depth	LR	SubS	ByTree	Gamma	Regα	Regλ	MinCW
32	**76.61%**	1141	13	0.0215	0.5749	0.6986	0.5589	0.00184	0.00259	5
12	76.46%	984	12	0.0145	0.6675	0.6946	0.0799	0.00503	0.00455	7
24	76.40%	1165	11	0.0162	0.7831	0.6403	0.00413	0.00481	0.00478	9
13	76.29%	995	12	0.0277	0.6345	0.6829	0.3999	0.00011	0.00567	6
18	76.29%	915	13	0.0121	0.6784	0.5958	0.1612	0.00491	0.1732	8

Top 5 trials from the hyperparameter search with corresponding parameters and validation accuracies. Bold marks the highest validation accuracy (trial 32).

manual feature engineering required by classical approaches [31, 37, 17, 32]. We explore architectures across two spatial granularities: pixel-level models that operate on per-pixel temporal sequences, and patch-level models that additionally exploit local spatial structure. To ensure unbiased evaluation across all architectures, all pixels belonging to a given field were assigned exclusively to either the training, validation, or test set, preventing spatial information leakage between splits.

3.2.1 Pixel-Level Architectures

At the pixel level, each observation is an individual pixel characterized by 60 temporal features spanning the 2017 growing season: six spectral bands and vegetation indices (B2, B6, B11, B12, EVI, Hue) observed across 10 monthly composites. Each pixel is linked to a unique field identifier and labeled according to the field crop type across five classes: wheat, barley, canola, small-grain grazing, and lucerne/medics. Two consequences of this design should be kept in mind. First, pixels within a field are highly spatially autocorrelated and all inherit the same field label (including mixed pixels along field boundaries), so although the pixel-level models are trained on millions of observations, the number of statistically independent units is closer to the number of fields than the number of pixels, and the pixel models' apparent data volume overstates the effective training size. Second, to prevent leakage we perform all train/validation/test splitting at the field level, so no field's pixels straddle two partitions; this keeps evaluation honest at the field scale even though training treats pixels as independent. Pixel-level predictions are subsequently aggregated to the field level via majority voting:

$$\hat{y}_f = \mathrm{mode}(\{\hat{y}_i \mid i \in f\})$$

For TabNet models we applied z-score standardization; CNN-based models were trained on raw reflectance values, allowing convolutional layers to learn directly from temporal magnitude patterns.

CNN + BiLSTM Ensemble with Focal Loss.

To capture the intricate temporal dynamics of crop growth, we developed a hybrid architecture combining 1D Convolutional Neural Networks (CNNs) with Bidirectional Long Short-Term Memory (BiLSTM) networks (Figure 7).

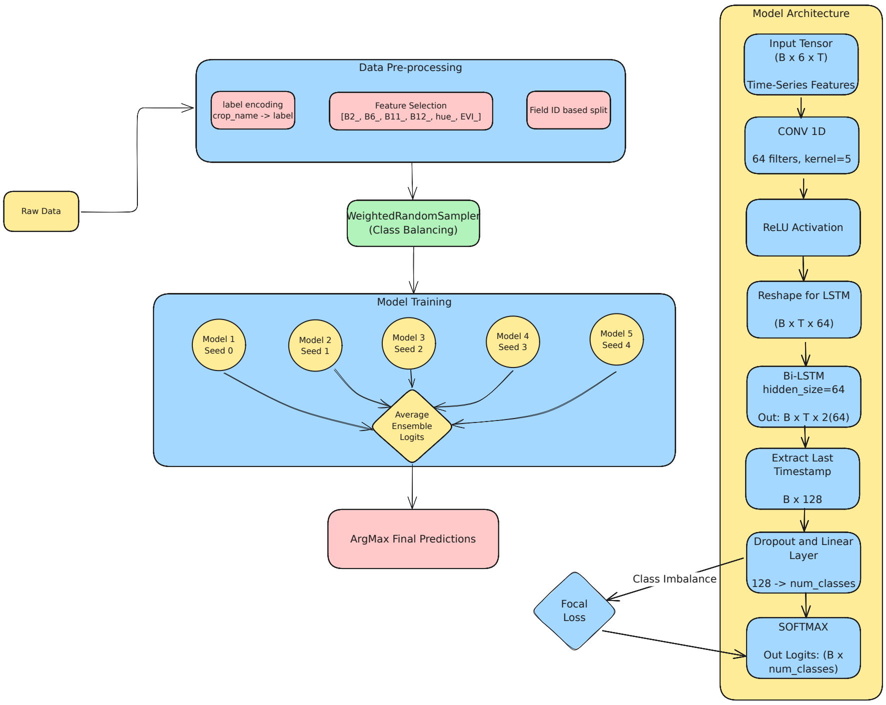

Fig. 7: Architecture of the CNN + BiLSTM model.

1D convolutional temporal feature extraction is combined with sequential modeling through a BiLSTM layer.

The model processes an input tensor of shape $(B, 6, T)$. A 1D convolutional layer with 64 filters and kernel size 5 is applied along the temporal axis, followed by ReLU activation and tensor permutation to $(B, T, 64)$ for compatibility with the recurrent layer. A bidirectional LSTM with hidden size 64 then captures temporal dependencies in both forward and backward directions, yielding an output of shape $(B, T, 128)$. The representation at the final timestep passes through dropout and a fully connected layer, with class probabilities computed via softmax:

$$P(y_i = c \mid \mathbf{x}_i) = \frac{e^{z_{i,c}}}{\sum_{c'=1}^{C} e^{z_{i,c'}}}$$

To address class imbalance we used Focal Loss:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)$$

where α_t is a per-class weight set to the inverse class frequency and $\gamma = 2.0$ is the focusing parameter, so class imbalance is handled directly in the loss rather than by resampling. To reduce variance, five identical models were trained with different random seeds and their logits averaged:

$$\text{Logits}_{\text{ensemble}} = \frac{1}{N}\sum_{k=1}^{N}\text{Logits}^{(k)}$$

Appendix D isolates the separate contributions of the five-seed ensemble and the weighted-focal objective (Tables 7 and 8); both affect holdout accuracy only at the margin.

TabNet with Sparsemax Feature Masks.

TabNet treats satellite observations as structured tabular inputs (each pixel's repeated measurements organized as a flat feature vector) and learns through iterative attention-based feature selection rather than temporal ordering. At each of n_{steps} decision steps, an Attentive Transformer produces a sparse feature mask via *sparsemax*,

$$\mathbf{M}[i] = \text{sparsemax}\big(\mathbf{P}[i-1]\odot h_i(\mathbf{a}[i-1])\big), \qquad \mathbf{P}[i] = \prod_{j=1}^{i}\big(\gamma - \mathbf{M}[j]\big),$$

where h_i is a trainable transform of the previous step's processed features $\mathbf{a}[i-1]$ and the prior scale $\mathbf{P}$ enforces the relaxation γ that discourages re-selecting the same features across steps; the masked features $\mathbf{M}[i]\odot\mathbf{f}$ are then handled by a Feature Transformer. This lets the model focus on different feature subsets at different depths without relying on explicit temporal structure [34, 33], and this sparsemax feature-mask mechanism is precisely what the L-TAE-S (below) grafts onto a temporal-attention backbone.

The network used $n_d = n_a = 64$, 2 independent and 2 shared feature-transformer blocks, relaxation parameter $\gamma = 1.5$, and 5 decision steps. Training used the Adam optimizer with a step-decay learning-rate schedule, batch size 1024, virtual batch size 128, and early stopping monitored on a field ID-stratified validation set.

An ensemble of five TabNet models, each initialized with a distinct random seed (42, 101, 202, 303, 404), averaged softmax outputs to yield final class probabilities:

$$\hat{\mathbf{p}}_{\text{ensemble}} = \frac{1}{5}\sum_{i=1}^{5}\text{Softmax}\left(\mathbf{z}^{(i)}\right), \qquad \hat{y} = \arg\max_{c}\,\hat{p}_{\text{ensemble},c}$$

A practical advantage of TabNet is interpretability: the sparse decision masks can be inspected post hoc to identify which spectral bands or vegetation indices most influenced each prediction [33], an instance of the post-hoc explainability increasingly emphasized in remote sensing [46].

Lightweight Temporal Attention Encoder (L-TAE).

The L-TAE applies self-attention directly to the temporal sequence of per-pixel reflectance vectors, learning which observations in the season are most informative

[35]. Given the embedded sequence $\{\mathbf{z}_t\}_{t=1}^T$, a set of learned master query vectors attend over time; for a single head the output is a temporally weighted sum

$$\mathbf{o} = \sum_{t=1}^{T} \alpha_t \, \mathbf{v}_t, \qquad \alpha_t = \frac{\exp\big(\mathbf{q}^\top \mathbf{k}_t / \sqrt{d}\big)}{\sum_{t'=1}^{T} \exp\big(\mathbf{q}^\top \mathbf{k}_{t'} / \sqrt{d}\big)},$$

where $\mathbf{k}_t, \mathbf{v}_t$ are key/value projections of $\mathbf{z}_t$, $\mathbf{q}$ is the learned query, and d is the head dimension. Channel grouping keeps the encoder lightweight. We evaluated L-TAE both at the pixel level and at the field level (averaging pixel sequences within a field before encoding), and trained a five-seed ensemble.

Pixel-Level Transformer Encoder.

As a dense-attention reference point we evaluated a standard Transformer encoder applied to the per-pixel reflectance sequence [19, 18]. After the same linear band embedding and sinusoidal positional encoding used for the L-TAE (with the true month indices, so the May–June gap is preserved), the embedded sequence $\mathbf{Z} = [\mathbf{z}_1, \ldots, \mathbf{z}_T]$ passes through three pre-norm multi-head self-attention layers, each computing

$$\text{head}_i = \text{softmax}\Big(\frac{(\mathbf{Z}W_i^Q)(\mathbf{Z}W_i^K)^\top}{\sqrt{d_k}}\Big)\mathbf{Z}W_i^V, \qquad \text{MHSA}(\mathbf{Z}) = \big[\,\text{head}_1 \,\|\cdots\|\, \text{head}_8\,\big]\, W^O,$$

with $d_{\text{model}} = 128$, 8 heads, and feed-forward width 256; unlike the L-TAE's single learned query, every time step here attends to every other. The encoded time steps are mean-pooled and passed to a two-layer classification head. This makes it the most expressive attention model we evaluate; yet, unlike the L-TAE-S below, it contains no feature-sparsity mechanism. It therefore serves as a control that separates attention *capacity* from feature *sparsity*. We trained it at the pixel level as a five-seed ensemble under the same weighted-focal-loss objective and field-aggregation protocol as the other temporal models.

Sparse-Attention L-TAE (L-TAE-S).

To test whether a tree-like, sparse inductive bias can be grafted onto the temporal attention encoder, we implemented a sparse-attention variant (L-TAE-S) that inserts a TabNet-inspired channel-selection gate before the attention heads (Figure 8). For each of the $n_{\text{head}} = 16$ heads, a per-head linear projection of the embedded sequence ($d_{\text{model}} = 128$) is passed through *sparsemax* [36] to produce a sparse, instance-dependent mask over the embedding channels, so each head attends to a small learned subset of features (with key dimension $d_k = 8$). The TabNet-style relaxation prior uses $\gamma = 1.5$ and the sparsity weight is $\lambda = 10^{-3}$; the resulting masks (which can be time-varying) are differentiable and interpretable, mirroring TabNet's sparsemax feature masks [34] while retaining the temporal-attention backbone of the L-TAE [35]. Formally, for head h and embedded, positionally-encoded time step $\mathbf{z}_t \in \mathbb{R}^E$ (with $E = d_{\text{model}}$), the gate produces a sparse channel mask $\mathbf{m}_{h,t}$ and a masked embedding

$\tilde{\mathbf{z}}_{h,t}$,

$$\mathbf{m}_{h,t} = \text{sparsemax}\big(\mathbf{P}_{h,t} \odot \text{BN}(W_h \mathbf{z}_t)\big), \qquad \tilde{\mathbf{z}}_{h,t} = \mathbf{m}_{h,t} \odot \mathbf{z}_t, \qquad \mathbf{P}_{h+1,t} = \mathbf{P}_{h,t} \odot \max(\gamma - \mathbf{m}_{h,t},\, 0),$$

initialized at $\mathbf{P}_{1,t} = \mathbf{1}$; head h then forms its keys and values from $\tilde{\mathbf{z}}_{h,t}$ before applying the L-TAE attention above. Because *sparsemax* projects onto the probability simplex, most channels of $\mathbf{m}_{h,t}$ are exactly zero, and the relaxation prior $\mathbf{P}_{h,t}$ (coefficient γ) discourages successive heads from re-selecting the same channels; a penalty $\lambda \sum_{h,t} \|\mathbf{m}_{h,t}\|_1$ adds mild additional pressure. (We use *sparsemax*; the α-entmax family [20] generalizes it with a tunable sparsity level and is a natural alternative we leave to future work.) We evaluate L-TAE-S as a first-class model at both the pixel and field level (a five-seed ensemble trained under the same weighted-focal-loss objective and field-aggregation protocol as the other temporal networks) and find that this single architectural change makes it one of the most transferable models in the study (Tables 3–4; Section 4.4).

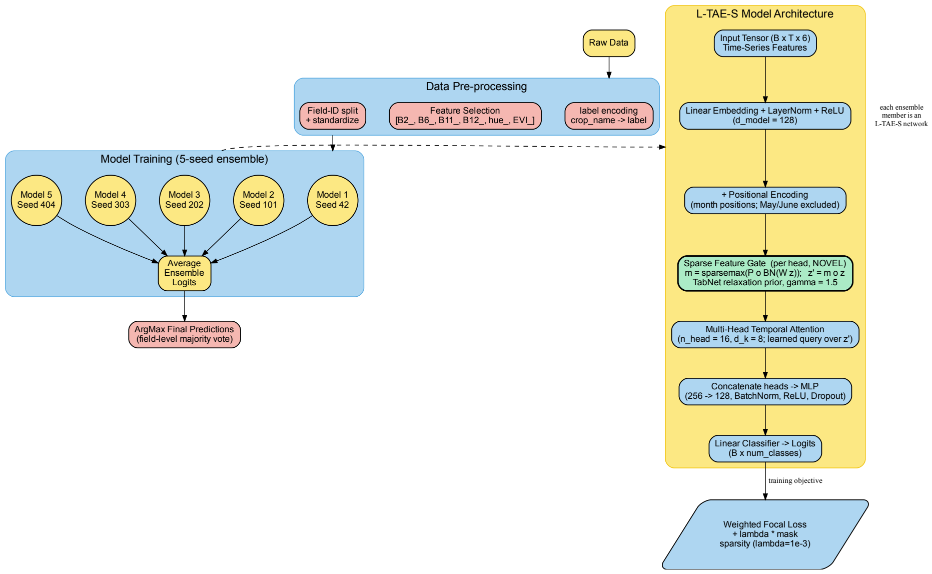

Fig. 8: Architecture of the sparse-attention L-TAE-S.

Each ensemble member embeds the per-pixel Sentinel-2 time series, adds month-aware positional encoding, and, before the multi-head temporal attention, applies a TabNet-style *sparse feature gate* (highlighted): a per-head `sparsemax` mask selects a small subset of embedding channels, with a relaxation prior ($\gamma = 1.5$) discouraging heads from re-selecting the same channels. This single block, which grafts a tree-like sparse feature-selection inductive bias onto the attention backbone, is the only architectural difference from the L-TAE. Five seeds are ensembled and aggregated to field level.

Temporal Convolutional Network (TempCNN).

TempCNN classifies a pixel from stacked one-dimensional convolutions applied along the temporal axis, each capturing local phenological motifs of increasing receptive field [37]. A convolutional layer computes

$$h_t^{(l)} = \mathrm{ReLU}\Big(\mathrm{BN}\left(\textstyle\sum_{\tau=-k}^{k} W_\tau^{(l)}\, h_{t+\tau}^{(l-1)} + b^{(l)}\right)\Big),$$

with batch normalization (BN) and dropout for regularization, followed by a fully connected softmax head. As with L-TAE, we evaluated pixel- and field-level variants and ensembled across five seeds.

3.2.2 Patch-Level Architectures

To exploit local spatial structure, we transitioned to a patch-level analysis in which each field is tiled into 100 m square patches (≈10×10 pixels at the 10 m resolution), each retaining the full 10-month spectral history for every pixel (Figure 9). Fields smaller than one hectare are kept whole as a single patch, and grid patches not fully contained within a field are discarded to avoid mixed-boundary pixels. Because the median field (7–11 ha) is several times larger than a patch, most fields yield several patches, providing implicit data augmentation. Before entering the convolutional models, each patch is resampled to a fixed 128×128 grid; we note that this upsamples the native ∼10×10-pixel patches by roughly an order of magnitude, so the enlarged input reflects interpolation rather than additional spatial detail.

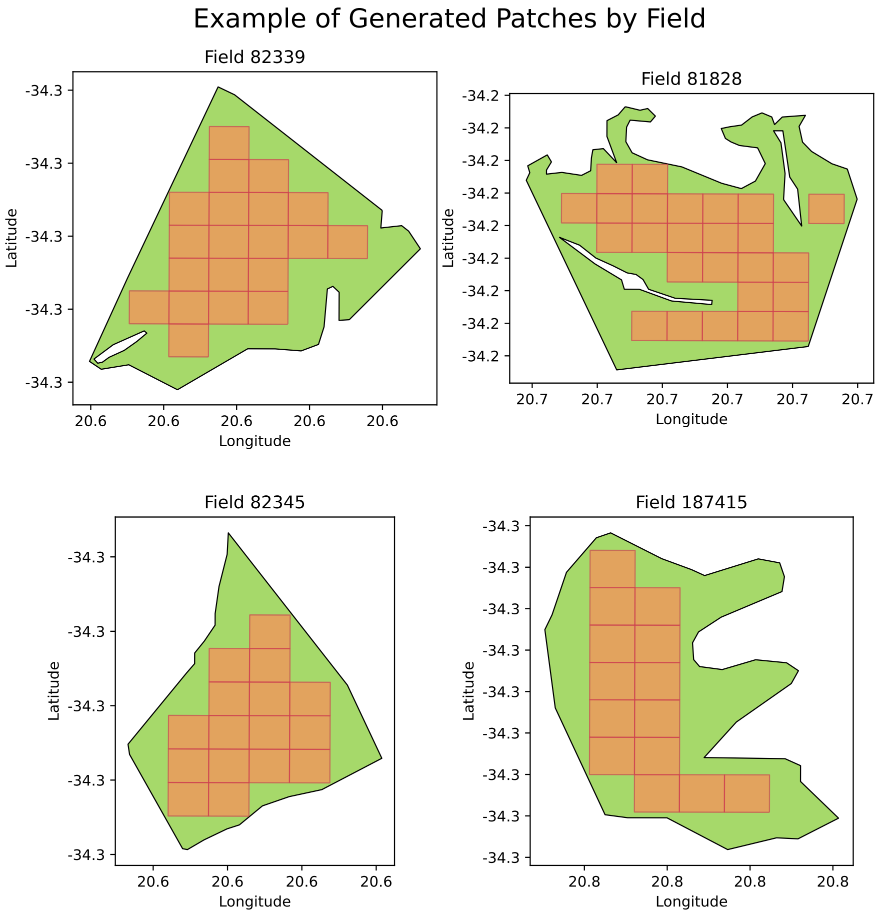

Fig. 9: Illustration of patch generation per field.

We investigated two distinct patch-based architectures.

Multi-Channel 2D CNN.

Each spectral band and monthly observation is treated as a separate input channel, giving a patch tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ ($H = W = 128$ after resampling, $C = 60 = 6$ features $\times$ 10 months). Two 2D-convolutional blocks (32 and 64 filters, 3×3 kernels), each followed by max-pooling, compute (for filter k):

$$Y_k[i, j] = \mathrm{ReLU}\left(\sum_{c=1}^{C}\sum_{m=-1}^{1}\sum_{n=-1}^{1} W_{k,c,m,n} \cdot X[i+m,\, j+n,\, c] + b_k\right)$$

The pooled feature map is flattened and projected by a dense hidden layer to $\mathbf{z} \in \mathbb{R}^{64}$ (ReLU), then passed to a fully connected classification head:

$$\hat{\mathbf{p}} = \mathrm{softmax}(W_{\mathrm{fc}}\, \mathbf{z} + b_{\mathrm{fc}})$$

The model is trained as a five-seed ensemble (identical architecture, distinct random seeds); per-patch class probabilities are averaged across the five seeds, and patch-level predictions are aggregated to a field label by majority vote, consistent with the field-level evaluation used throughout.

3D CNN.

A single 3D CNN treats each patch as a spatiotemporal volume $\mathbf{X} \in \mathbb{R}^{T\times H\times W\times C}$ ($T = 10$ months, $H = W = 128$, $C = 6$ bands) and applies three 3D-convolutional blocks (32, 64, and 128 filters, $3 \times 3 \times 3$ kernels) with spatial max-pooling, followed by global average pooling and a 128-unit ReLU dense layer producing $\mathbf{z}$, then a softmax classification head over the five crops:

$$\hat{\mathbf{p}} = \mathrm{softmax}(W_{\mathrm{fc}}\, \mathbf{z} + b_{\mathrm{fc}})$$

Class imbalance is handled directly in the loss via a weighted focal loss ($\gamma = 2.0$, per-class weight α_t set to inverse class frequency). By modelling volumetric relationships jointly across spectral bands, spatial dimensions, and time, 3D convolutions capture crop-specific texture and phenological trajectories that 2D methods treat independently. As with the temporal networks, the model is trained as a five-seed ensemble whose softmax outputs are averaged across seeds, and patch predictions are aggregated to a field label by majority vote (Figure 10).

3.3 Performance Metrics

Because the class distribution is severely imbalanced (Figure 1), plain accuracy is misleading: a classifier can score highly simply by predicting the majority class. We therefore adopt macro-averaged F1 ($\mathrm{F1}_m$), the unweighted mean of the per-class F1 scores, as our primary metric: it weights every crop equally and so directly penalizes failure on the minority cereals, which is exactly where models differ. We report three further metrics alongside it: Cohen's Kappa, chance-corrected agreement between predicted and true labels; weighted F1, the support-weighted mean of per-class F1; and cross-entropy (Xent), the field-level cross-entropy that served as the official scoring metric of the AI4FoodSecurity (South Africa) challenge [3] from which our dataset is drawn (lower is better; computed on hard crop-label predictions, Appendix B). Higher Kappa and F1 indicate better, more balanced classification. Formal definitions of all four metrics are given in Appendix B.

3.4 Quantifying uncertainty

Because our claims rest on between-model differences, we quantify two sources of variability. First, to capture sampling uncertainty in the finite holdout, we apply a

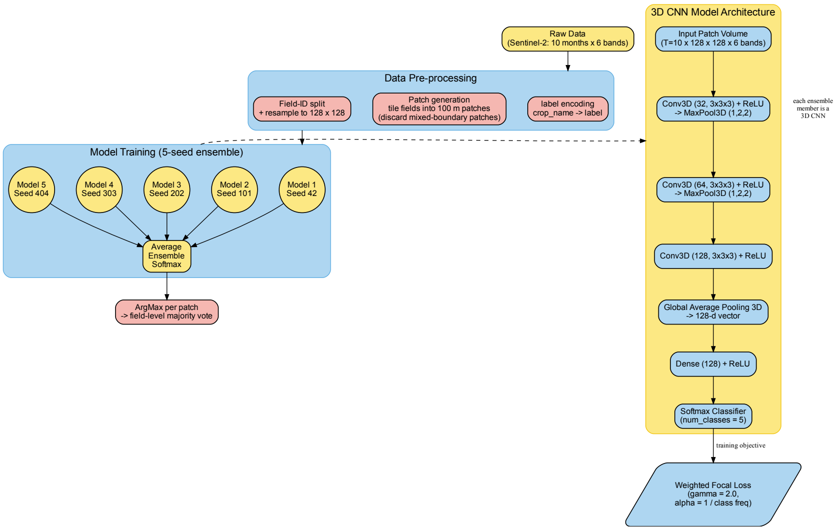

Fig. 10: Patch-level 3D CNN pipeline. Each field is tiled into 100 m patches and resampled to 128×128; a single multiclass 3D CNN is trained as a five-seed ensemble whose softmax outputs are averaged, and per-patch predictions are aggregated to a field label by majority vote.

nonparametric bootstrap over the holdout fields: for each model we resample its $\sim$2,415 per-field predictions with replacement $B = 10{,}000$ times and recompute macro-F1 (and per-class F1) on each replicate, reporting the 2.5th–97.5th percentile interval as a 95% confidence interval (CI) on its ST F1$_m$ (Table 3, Figures 11–12). For a between-model gap Δ we take the difference of the two models' independent bootstrap distributions and report its CI and a two-sided bootstrap p-value; treating the resamples as independent is conservative, since it ignores the positive correlation induced by the shared holdout, so any gap flagged significant is robustly so. Second, to capture optimization noise, we report the spread of ST F1$_m$ across the five random-seed members of the deep ensembles (Section 3.3 describes the ensembling): the per-seed standard deviation is 0.015–0.037 for the headline temporal models (L-TAE, TempCNN, CNN-BiLSTM at the pixel and field levels), so a single-seed run varies by up to ≈ 0.04 macro-F1, which is precisely why we report five-seed ensembles rather than individual runs throughout. Both sources are small relative to the ranking effects we report: every field-bootstrap CI spans only $\approx \pm 0.02$–0.03 macro-F1, so we treat ST F1$_m$ differences below ≈ 0.05 as not statistically distinguishable throughout.

4 Results

We evaluated every model under two protocols. *In-region field-wise validation* uses a field-wise train/validation/test split drawn entirely from the training tiles, this is the conventional way crop classifiers are benchmarked; we shorten this to "in-region". *Spatial transfer* (ST) applies the trained model to the spatially disjoint holdout tile (`34S_20E_259N`; 2,417 fields). All predictions are aggregated to the field level. Because the classes are heavily imbalanced (Section 3.3), we report macro-averaged F1 ($\mathrm{F1}_m$) as the primary metric, with Cohen's Kappa, weighted F1, and log-loss (cross-entropy) alongside.

4.1 In-region validation overstates performance and misranks models

Table 3 reports both protocols for all individual models, ranked by spatial-transfer macro-F1, with the spatial-transfer gap $\Delta = \mathrm{F1}_m^{\mathrm{ST}} - \mathrm{F1}_m^{\text{in-region}}$ shown alongside; Figure 11 visualizes the same data ordered by Δ. Two facts stand out. First, *every* model loses accuracy under spatial transfer, but the loss is highly uneven: it ranges from -0.04 (the field-level sparse-attention L-TAE-S) to -0.26 (CNN-BiLSTM at the pixel level). Second, this unevenness reorders the models. Under in-region validation the dense temporal pixel-level deep nets cluster at the top, with $\mathrm{F1}_m$ scores for L-TAE at 0.78, Transformer 0.78, L-TAE-S 0.77, TempCNN 0.77, CNN-BiLSTM 0.72. A reasonable practitioner choosing by in-region F1 would pick from this cluster. On the holdout tile the cluster fractures sharply: the two in-region leaders (L-TAE pixel and the Transformer encoder, $\Delta = -0.20$) drop to $\mathrm{F1}_m = 0.58$, TempCNN (-0.21) to 0.56, and CNN-BiLSTM (-0.26) to 0.46, this is among the lowest absolute scores of any model. Only L-TAE-S, the sparse-attention variant, holds its ground. The same in-region-to-holdout collapse befalls the 3D CNN patch ensemble (-0.20), the Multi-Channel 2D CNN (-0.18), and SMOTE-resampled stacking (-0.21).

The winners under spatial transfer are a different group entirely. The best single model is the TabNet pixel ensemble ($\mathrm{F1}_m = 0.60$, Kappa 0.54), which also posts the lowest holdout cross-entropy on the challenge's official metric (4.42) of any model, tied on macro-F1 by both sparse-attention L-TAE-S variants ($\mathrm{F1}_m = 0.60$; Section 4.4). The most *robust* models, those with the smallest gaps, are the sparse learners: the field-level L-TAE-S posts the single smallest gap of any model ($\Delta = -0.04$), just ahead of logistic regression (-0.05), pixel-level LightGBM (-0.06), and the pixel-level tree learners. That L-TAE-S sits at the top of the robustness ranking is itself evidence for our thesis. It is a deep temporal attention network, but with one twist: a learned gate restricts each attention head to a small subset of input channels rather than mixing all of them. That built-in selectivity, copied directly from TabNet, is precisely what gives it the cross-region stability the other deep nets lack. Notably, the robust `xr_fresh` classical ensembles in the upper-middle of the table (XGBoost, Voting, LightGBM, Stacking) are the models fed the automated `xr_fresh` features (Table 3, Features column): the combination of a rich automated feature representation with a tree learner transfers well, whereas the deep nets that learn from raw sequences do not. The practical implication is immediate: a user who picked by in-region $\mathrm{F1}_m$ would have landed on L-TAE, the Transformer encoder, or TempCNN (all 0.77–0.78 in-region), each of which drops to 0.56–0.58 on the holdout and falls below TabNet (0.60). CNN-BiLSTM, the model prior analysis of this dataset singled out [5], fares worse still under transfer ($\mathrm{F1}_m = 0.46$, among the lowest absolute scores in the benchmark).

Table 3: In-region versus spatial-transfer performance of all models.

Model	Level	Features	In-reg		Spatial transfer (ST)				
			F1$_m$	Δ	F1$_m$	95% CI	κ	wF1	Xent
TabNet	pixel	raw	0.73	−0.13	**0.60**	.58–.63	**0.54**	0.70	**4.42**
L-TAE-S	field	raw	0.64	**−0.04**	0.60	.58–.62	0.52	0.69	4.77
L-TAE-S	pixel	raw	0.77	−0.17	0.60	.57–.62	0.53	**0.70**	4.63
L-TAE	field	raw	0.67	−0.07	0.59	.57–.62	0.47	0.67	5.51
L-TAE	pixel	raw	**0.78**	−0.20	0.58	.55–.60	0.49	0.68	5.20
Transformer	pixel	raw	**0.78**	−0.20	0.58	.55–.60	0.50	0.69	4.94
XGBoost	field	`xr_fresh`	0.70	−0.13	0.56	.54–.59	0.47	0.68	4.89
TempCNN	pixel	raw	0.77	−0.21	0.56	.54–.59	0.49	0.68	5.09
Base LightGBM	pixel	`xr_fresh`	0.62	−0.06	0.56	.54–.59	0.47	0.66	4.85
Logistic Regression	pixel	`xr_fresh`	0.61	−0.05	0.56	.54–.59	0.47	0.67	4.93
LightGBM	field	`xr_fresh`	0.68	−0.12	0.56	.53–.58	0.45	0.66	4.99
Voting	pixel	`xr_fresh`	0.65	−0.09	0.56	.53–.58	0.48	0.66	4.85
Random Forest	pixel	`xr_fresh`	0.65	−0.10	0.55	.52–.57	0.45	0.65	5.07
Base XGBoost	pixel	`xr_fresh`	0.64	−0.09	0.54	.52–.57	0.47	0.66	5.01
TempCNN	field	raw	0.67	−0.14	0.54	.51–.56	0.43	0.64	6.07
Stacking	field	`xr_fresh`	0.64	−0.11	0.53	.51–.56	0.48	0.67	5.32
CNN-BiLSTM	field	raw	0.64	−0.14	0.51	.48–.53	0.33	0.54	7.86
3D CNN	patch	raw	0.70	−0.20	0.50	.48–.53	0.29	0.54	7.98
SMOTE Stacked	field	`xr_fresh`	0.69	−0.21	0.49	.46–.51	0.43	0.63	5.37
CNN-BiLSTM	pixel	raw	0.72	−0.26	0.46	.44–.48	0.32	0.56	7.73
Multi-Channel CNN	patch	raw	0.63	−0.18	0.45	.43–.47	0.30	0.54	7.62
TabNet Temporal	field	raw	0.62	−0.19	0.43	.41–.46	0.31	0.56	7.35
TabNet	field	`xr_fresh`	0.51	−0.15	0.37	.34–.40	0.25	0.53	6.13

In-region (field-wise validation) versus spatial-transfer (ST) macro-F1 for every individual model, ranked by spatial-transfer macro-F1, the metric we argue model selection should use. In-region macro-F1 and the gap Δ are shown alongside (Figure 11 orders the same data by Δ); ST Kappa, weighted F1 (wF1), and cross-entropy (Xent, the AI4FoodSecurity challenge's official field-level scoring metric [3], lower is better) complete the picture. The **95% CI** column gives the percentile bootstrap interval on ST F1$_m$ from 10,000 resamples of the 2,415 holdout fields (Section 3.4). Bold marks the best value in each metric column. The best transferer is a TabNet pixel ensemble; the highest in-region scores (L-TAE and Transformer pixel, both 0.78; L-TAE-S and TempCNN pixel, both 0.77) do not yield the highest spatial-transfer scores, and CNN-BiLSTM, singled out by prior analysis of this dataset [5], suffers the largest drop of any model ($\Delta = -0.26$ at the pixel level). The *Features* column marks whether a model is fed the automated `xr_fresh` time-series statistics (174 features) or raw reflectance; the `xr_fresh` models are the classical learners: the four *Base* models on per-pixel statistics, the field ensembles on field-averaged statistics.

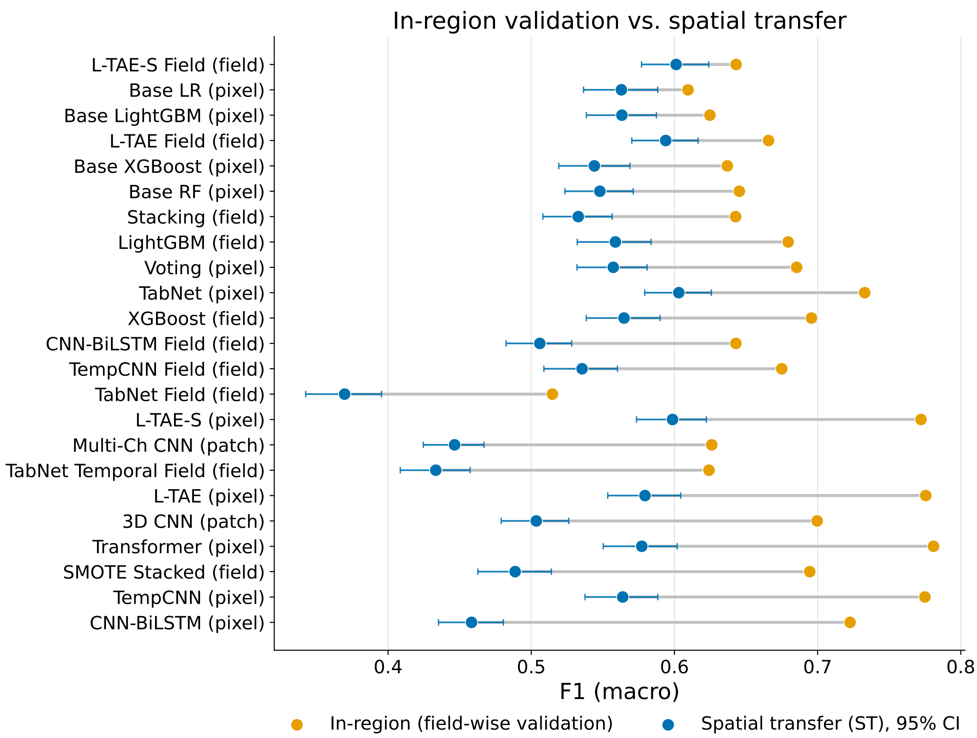

Fig. 11: In-region versus spatial-transfer macro-F1 by model, sorted by gap.

In-region performance is from field-wise validation. Dense temporal/patch deep nets (top) suffer the largest degradation; tree-based, linear, and field-aggregated models (bottom) transfer with the smallest gaps. Whiskers on the spatial-transfer points are 95% bootstrap confidence intervals over the holdout fields (Section 3.4).

The field bootstrap (Section 3.4) confirms which of these differences are statistically real and which are not. The thesis-critical gaps survive: the best transferers beat CNN-BiLSTM by a wide, significant margin (TabNet over CNN-BiLSTM pixel, $\Delta = 0.14$, 95% CI $[0.11, 0.18]$, $p < 0.001$; the same holds for both L-TAE-S variants), so the ranking inversion is not an artifact of holdout noise. At the very top, by contrast, the leaders are statistically indistinguishable: TabNet, the field- and pixel-level L-TAE-S, and the field-level L-TAE all fall within 0.59–0.60 and their pairwise gaps are not significant (TabNet vs. L-TAE-S, $\Delta = 0.004$, $p = 0.80$; TabNet vs. pixel L-TAE, $\Delta = 0.024$, $p = 0.19$). We therefore do not claim a single best model; we claim that a sparse-feature-selection group occupies the top of the spatial-transfer ranking and that the dense temporal nets prized by in-region validation fall significantly below it.

4.2 Pixel versus field level: aggregation as variance reduction

Several architectures were run at both the pixel level and the field level (averaging the per-pixel time series within each field before classification). Table 4 and Figure 12 show that field-level aggregation acts as a variance-reduction regularizer for the *dense* temporal networks: L-TAE improves from $\mathrm{F1}_m = 0.58$ (pixel) to 0.59 (field) while its gap shrinks from -0.20 to -0.07, and CNN-BiLSTM, the worst pixel-level transferer, recovers most of all when field-trained, its gap halving from -0.26 to -0.14 ($\mathrm{F1}_m$ $0.46 \rightarrow 0.51$). TempCNN is a partial exception: its gap narrows from -0.21 to -0.14, but its ST $\mathrm{F1}_m$ slips slightly ($0.56 \rightarrow 0.54$), so the gap narrowing reflects a drop in the in-region ceiling more than a holdout gain. The sparse-attention L-TAE-S follows the same dense-backbone pattern despite its TabNet-style channel gating: its gap shrinks from -0.17 (pixel) to -0.04 (field, the smallest of any model) while ST $\mathrm{F1}_m$ holds at 0.60; so the temporal backbone, not the sparse gate, governs the response to aggregation. Averaging pixels within a field cancels region-specific pixel noise before the model sees it, lowering the in-region ceiling and, for most dense models, improving transfer. TabNet is the informative exception: aggregating its raw inputs to the field level *hurts* ($\mathrm{F1}_m$ falls from 0.60 to 0.43), because its robustness already derives from sparse feature selection and field-level training starves it of the per-pixel observation volume it exploits (millions of pixels versus a few thousand fields). The tree ensembles transfer well at either level. In short, dense models need aggregation to generalize; sparse, tree-like models do not.

A second comparison is embedded in the same table. The tree learners are run at both granularities on the *same* `xr_fresh` representation: the *Base* models (Logistic Regression, Random Forest, LightGBM, XGBoost) on per-pixel statistics, and the field ensembles (XGBoost, LightGBM) on field-averaged statistics, so their pixel→field change isolates the effect of aggregation alone, with the feature representation held fixed. That change is essentially neutral under spatial transfer: XGBoost moves from $\mathrm{F1}_m = 0.54$ (pixel) to 0.56 (field) and LightGBM stays at 0.56, even though aggregation raises the in-region score (e.g. XGBoost $0.64 \rightarrow 0.70$) and therefore widens the gap. Aggregation thus adds little for the already-sparse tree learners; the marginal value of the `xr_fresh` representation itself, as opposed to the aggregation level, is taken up separately in Section 4.4. TabNet, run at the field level with both raw field-averaged and `xr_fresh` inputs, behaves consistently: moving its raw inputs from pixel to field level costs $\mathrm{F1}_m$ $0.60 \rightarrow 0.43$ (the aggregation/observation-volume penalty), and the field `xr_fresh` variant is lower still at 0.37. So for TabNet aggregation does not help, consistent with the broader finding that already-sparse models gain nothing from field-level averaging.

4.3 Data efficiency under spatial transfer

To probe how much labeled data the transferable models actually require, we retrained a representative set on random subsets of 75%, 50%, and 25% of the training fields and re-evaluated on the holdout tile (Figure 13). The most robust models are also the most data-efficient: TabNet retains essentially full holdout accuracy down to a quarter of the fields ($\mathrm{F1}_m$ = 0.60, 0.62, 0.62, 0.59 at 100/75/50/25%), and logistic regression is similarly flat (0.56–0.60); the sparse-attention L-TAE-S tracks them ($\mathrm{F1}_m = 0.60, 0.61, 0.56, 0.58$ at 100/75/50/25%), whereas the dense pixel L-TAE degrades more steeply (0.58 to 0.57). Holdout accuracy is therefore limited far more by spatial transfer than by training-set size for the robust models: halving the labels costs them little.

Table 4: Pixel- versus field-level spatial transfer.

	Pixel		Field		
Architecture	**ST F1$_m$**	Δ	**ST F1$_m$**	Δ	**Field input**
L-TAE-S	**0.60**	−0.17	**0.60**	**−0.04**	raw-avg
L-TAE	0.58	−0.20	0.59	−0.07	raw-avg
XGBoost	0.54	−0.09	0.56	−0.13	`xr_fresh`
LightGBM	0.56	**−0.06**	0.56	−0.12	`xr_fresh`
TempCNN	0.56	−0.21	0.54	−0.14	raw-avg
CNN-BiLSTM	0.46	−0.26	0.51	−0.14	raw-avg
TabNet	**0.60**	−0.13	0.43	−0.19	raw-avg

Spatial-transfer (ST) macro-F1 (and gap Δ) at pixel versus field level for architectures evaluated at both granularities, ranked by field-level ST F1$_m$. Bold marks the best value in each column. The pixel input is raw reflectance for the deep models and per-pixel `xr_fresh` statistics for the tree learners; the *Field input* column gives the field-level representation, either `xr_fresh` (field-averaged) for the trees or the field-averaged raw time series (raw-avg) for the deep models. For the tree learners (XGBoost, LightGBM) the pixel→field change is therefore purely an aggregation change at a fixed `xr_fresh` representation. Field aggregation helps the dense temporal nets and is roughly neutral for the tree models, but degrades TabNet. The field bootstrap (Section 3.4; 95% CIs shown in Figure 12) confirms that the two largest aggregation effects are statistically significant, CNN-BiLSTM improves ($\Delta = +0.05$, $p = 0.005$) and TabNet degrades ($\Delta = -0.17$, $p < 0.001$), whereas the pixel↔field changes for L-TAE, TempCNN, and the tree learners fall within the $\approx \pm 0.02$–0.03 CIs and are not significant.

4.4 The `xr_fresh` representation: value and redundancy

The classical learners that transfer best in Table 3 are precisely those fed the automated `xr_fresh` statistics, yet that table alone cannot say whether their robustness comes from the representation or from the tree learners themselves: every raw-reflectance model in it is a deep net and every `xr_fresh` model is a tree, so representation and model family are confounded. To separate them we ran a controlled comparison in which only the representation changes. A single XGBoost learner (the manuscript's pixel-level configuration), with the FID-wise splits, preprocessing, and field-level mean pooling all held fixed, was trained once on raw monthly reflectance (six bands × ten months) and once on per-pixel `xr_fresh` statistics, each evaluated both in-region and under spatial transfer (Table 5). The two representations are essentially indistinguishable under transfer: raw reflectance reaches F1$_m$ = 0.59 and `xr_fresh` 0.56 on the holdout tile (0.73 vs. 0.72 in-region), with `xr_fresh` marginally lower on macro-F1 but marginally better on field-level cross-entropy (0.82 vs. 0.85) and equal on weighted-F1 and κ. The `xr_fresh` representation therefore yields no measurable spatial-transfer gain over raw reflectance for the trees; its value is operational and interpretive rather than an accuracy advantage. Operationally, it compresses each pixel's ten-month, six-band series into a single compact vector that a tree fits in seconds to minutes, versus tens of minutes to hours per seed for the dense deep nets (Appendix C). Interpretively, unlike an anonymous vector of monthly reflectances each `xr_fresh` feature carries an explicit, agronomically readable meaning, such as the day-of-year of a band's seasonal maximum, its temporal autocorrelation, or the skewness

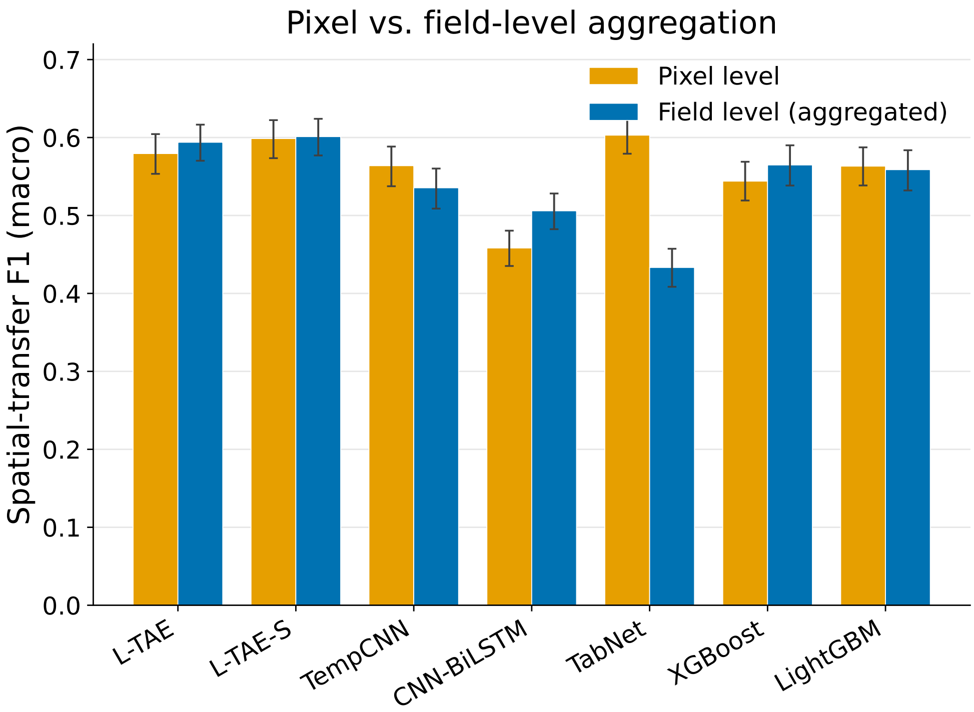

Fig. 12: Spatial-transfer macro-F1 at pixel versus field level.

Field-level aggregation recovers transfer for dense temporal models (L-TAE, CNN-BiLSTM) but collapses the sparsity-reliant TabNet. Error bars are 95% bootstrap confidence intervals over the holdout fields (Section 3.4).

of its trajectory, so feature-importance rankings translate directly into phenological statements about how the crops separate.

Given that we adopt the `xr_fresh` representation for these operational and interpretive reasons rather than for any accuracy edge, a natural follow-on concern is whether its 174 dimensions are themselves a liability through feature redundancy. We tested two forms of explicit dimensionality control. A *sparse-attention* L-TAE variant (L-TAE-S, TabNet-style channel gating via sparsemax) modestly improved transfer over the dense L-TAE at full data (0.60 vs. 0.58 at the pixel level and 0.60 vs. 0.59 at the field level; Tables 3 and 4) and at matched reduced data (0.61 vs. 0.57 at 75% of fields), consistent with *soft*, learned sparsity providing a modest but consistent transfer benefit. L_2 *regularization* of the tree models changed ST $\mathrm{F1}_m$ negligibly (within ± 0.01). Together these show that the high feature dimensionality is not the binding constraint: soft, learned sparsity helps a little, blunt magnitude penalties do not, and the dominant limitation remains spatial transfer.

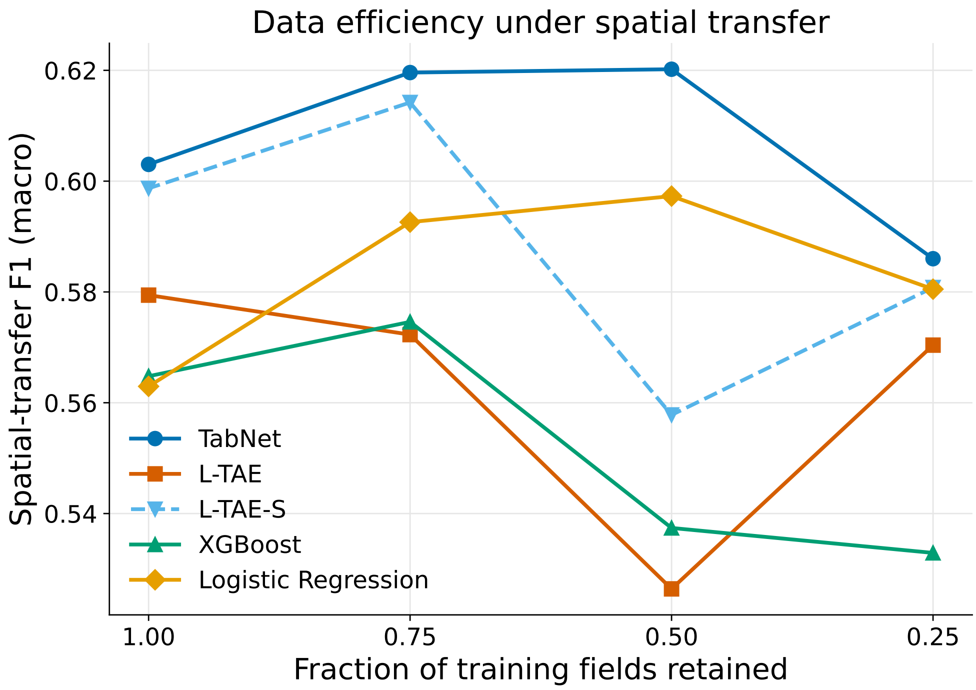

Fig. 13: Spatial-transfer macro-F1 versus fraction of training fields.

Models are retrained on random subsets of 75%, 50%, and 25% of the training fields and re-evaluated on the holdout tile. Robust models (TabNet, logistic regression, and the sparse-attention L-TAE-S) are nearly flat.

4.5 Per-class performance and persistent confusions

Figure 14 breaks down accuracy crop by crop for four representative models on the holdout region: the best overall transferer (TabNet), our sparse-attention network (L-TAE-S), its standard counterpart without the sparsity gate (L-TAE), and CNN-BiLSTM, whose accuracy fell the most from the training region to the holdout. The errors are highly uneven across crops. All three robust models classify the dominant crop lucerne/medics and the distinctive canola well (TabNet F1 0.88 and 0.81; L-TAE-S 0.87 and 0.76; L-TAE 0.84 and 0.66), but every model struggles with the cereals and with small-grain grazing: even TabNet, the best transferer, reaches only 0.55 on barley, 0.48 on wheat, and 0.30 on small-grain grazing. The L-TAE → L-TAE-S contrast is informative. Adding the sparsity gate to the same temporal backbone modestly improves the easier classes (canola 0.66 → 0.76, lucerne/medics 0.84 → 0.87) while leaving the hard cereals essentially unchanged (barley 0.50 → 0.50, wheat 0.48 → 0.47); the average-F1 gain therefore reflects sharper performance on the easy classes, not a breakthrough on the difficult ones. CNN-BiLSTM, by contrast, loses accuracy on *every* class — including the easy ones (lucerne/medics drops to 0.71, canola to 0.65, and the cereals collapse to roughly 0.28 each) — showing that its failure is a general inability to transfer to the new region, not a problem caused by the minority

Table 5: Raw reflectance versus **xr_fresh** for a fixed tree learner.

Representation	**Split**	**F1**$_m$	**wF1**	κ	**Xent**
Raw reflectance	In-region	0.73	0.80	0.71	0.65
xr_fresh	In-region	0.72	0.76	0.68	0.74
Raw reflectance	Spatial transfer	0.59	0.69	0.50	0.85
xr_fresh	Spatial transfer	0.56	0.69	0.50	0.82

A single XGBoost learner (the manuscript's pixel-level configuration) trained on raw monthly reflectance (six bands × ten months) versus per-pixel xr_fresh statistics, with identical FID-wise splits, preprocessing, and field-level mean pooling; only the input representation differs. In-region uses held-out fields from the two training tiles; spatial transfer uses the disjoint holdout tile (34S_20E_259N). The two representations are essentially indistinguishable under transfer, confirming that xr_fresh's value for the trees is operational (compactness and training speed; Appendix C) rather than a transfer-accuracy gain.

crops alone. The persistent confusion between wheat and barley is consistent with their spectral overlap (Figure 4) and appears across every model we examined.

5 Discussion

Our results demonstrate that, in this setting, the conclusion one draws about which model is "best" depends almost entirely on whether evaluation is in-region or under spatial transfer, and that a model's transfer behaviour is predictable from its inductive bias.

Why tree-like inductive bias transfers.

The single clearest pattern in our results is that transfer tracks *inductive bias*, not model family or capacity per se. Figure 15 groups every model by inductive bias and plots its spatial-transfer gap. The ordering is stark: linear models lose the least (mean $\Delta = -0.05$), tree-based models little (mean -0.11), the sparse-attention L-TAE-S sits essentially with the trees (mean -0.11 across its pixel and field-aggregated variants), while dense temporal/patch deep nets lose the most (mean -0.21) and SMOTE-augmented stacking is similarly fragile (-0.21). Random Forest and gradient-boosted trees have been the remote-sensing workhorse for two decades not because they always produce the highest in-region accuracy (across our benchmark and the broader literature, dense deep nets often do), but because they reliably survive when the model is applied to a region it was not trained on, which is the situation operational crop mapping actually faces (see [14], §6.1, for an early documentation of this effect for Random Forest, and [30] for the broader machine-learning view). Our results explain why mechanistically. A tree decides one feature at a time, choosing the few features that matter for each split and ignoring the rest. That sparsity, what statisticians have long called feature selection, is precisely what keeps the model from latching onto the region-specific quirks of phenology, soil, and atmosphere that pull richer models

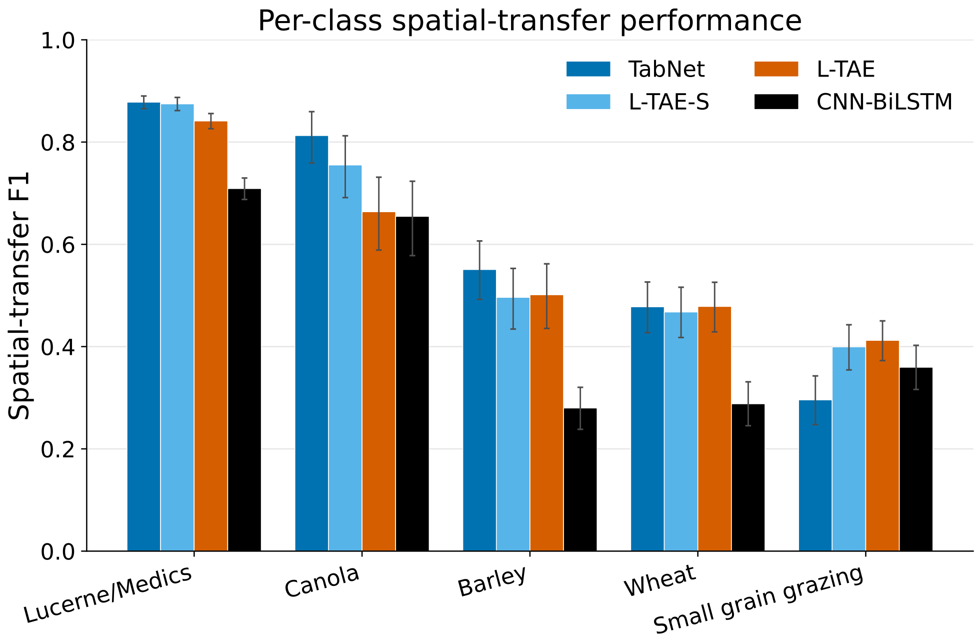

Fig. 14: Per-class accuracy (F1) on the holdout region for four representative models.

Each group of bars shows the four models' accuracy on one crop class. TabNet, L-TAE-S, and L-TAE classify lucerne/medics and canola well; the cereals (barley, wheat) and small-grain grazing are where most of the errors concentrate. CNN-BiLSTM's accuracy drops on every class, including the easy ones — a sign that its failure is a general transfer problem, not just an imbalance problem. Error bars are 95% bootstrap confidence intervals over the holdout fields (Section 3.4); they widen on the minority classes (canola, small-grain grazing), whose per-class F1 is estimated from fewer fields.

off course. Independent corroboration arrives from a very different setting: Pankajakshan et al. [49] show on hyperspectral imagery that a 3D-CNN collapses from 97–99% in-region accuracy to "very poor" under cross-sensor, cross-region transfer while a sparse SVM transfers; the architectures and sensors differ, but the mechanism is the same. Convergent evidence appears in other Sub-Saharan settings: Rustowicz et al. [38] report that Random Forest matches or exceeds a dense CNN-LSTM on Ghana smallholder fields where labels are scarce, even as dense models dominate on data-rich Germany; and Jin et al. [12] document a 16-point holdout drop (79% Tanzania to 63% Kenya) when their maize model is moved across the East African landscape. The direction and magnitude of these drops are consistent with our own. The same tree-favouring ordering holds under a far larger, cross-continental shift: transferring crop classifiers from the United States Cropland Data Layer to the Hexi Corridor of north-west China, Mai et al. [40] find bagged and gradient-boosted trees more robust than a more elaborate instance-reweighting transfer-boosting scheme, with zero-target-label transfer degrading to a macro-F1 near 0.45; their benchmark contains tree learners only, so it corroborates the tree-survives evidence rather than that dense-nets collapse. The best spatial-transfer model in our benchmark, TabNet, is a neural network built to behave like a tree: at each of its internal decision steps a small learned gate picks a handful of features and ignores the rest, the same one-feature-at-a-time principle that drives Random Forest and XGBoost [34]. TabNet therefore clusters with the trees, not with the dense sequence models, and inherits their transferability. By contrast, CNN-BiLSTM, TempCNN, L-TAE, and the plain Transformer encoder consume the whole temporal sequence at once, mixing every feature with every other; with no built-in feature selection they fit the training region's signal tightly and transfer poorly. Independent analysis of attention in satellite image time series finds that the dates such models learn to attend to are themselves contingent on the training crops and region [21], consistent with this region-overfitting. The Transformer encoder is a particularly clean control: although it deploys the most expressive attention of any model we test (full multi-head self-attention stacked over several layers, far richer than L-TAE's single learned query), it transfers no better than L-TAE ($F1_m = 0.58$, $\Delta = -0.20$ at the pixel level). Adding attention *capacity* thus does nothing for spatial transfer, whereas adding feature *selection* does: L-TAE-S and TabNet, the two models that explicitly select features before computing with them, are the best transferers. This isolates feature selection, not the attention mechanism, the temporal backbone, or model size, as the property that governs spatial transfer.

What the L-TAE-S experiment adds is that this is *not* a tree-versus-deep-learning trade-off. The same feature-selection step that drives tree splits can be added to any dense architecture: in L-TAE-S we place it just before the temporal attention; in TabNet it sits before the network's main computation; the same step could in principle be added to CNNs, recurrent networks, or other deep architectures. The architectural cost is a single block, and the transfer payoff is large: L-TAE-S posts the smallest spatial-transfer gap of any model in our benchmark and ties TabNet for the best absolute holdout accuracy. We read this as evidence that the next generation of operational crop classifiers does not need to choose between the expressive power of deep learning and the transfer-robustness of trees; the two can coexist when feature selection is built in explicitly rather than left to emerge by accident.

Two robustness levers.

We are careful not to overclaim a single tidy axis, because two caveats sharpen rather than weaken the story. First, it is *pixel-level* TabNet that wins: the field-aggregated TabNet variants are among the worst transferers (Table 3), because aggregation removes the per-pixel observation volume that the sparse model exploits. Second, dense temporal attention transfers well *when* applied at the field level: L-TAE's gap shrinks from $\Delta = -0.20$ at the pixel level to -0.07 at the field level, and our sparse-attention L-TAE-S shrinks further still, from -0.17 to -0.04 when field-aggregated, the smallest in-region-to-holdout gap of any model in the study (Table 4). These observations point to two distinct, partly substitutable pathways to transfer-robust models, and together they map the design space for operational remote-sensing classifiers. The first lever is built-in feature selection, the model itself picking a few features at a time (trees, TabNet, and now L-TAE-S via its sparse feature gate). This is *intrinsic* robustness, paid for at model-design time. The second lever is variance reduction imposed from outside the model via field-level aggregation (Section 4.2), which averages away region-specific pixel noise before the classifier sees it. This is *external* robustness,

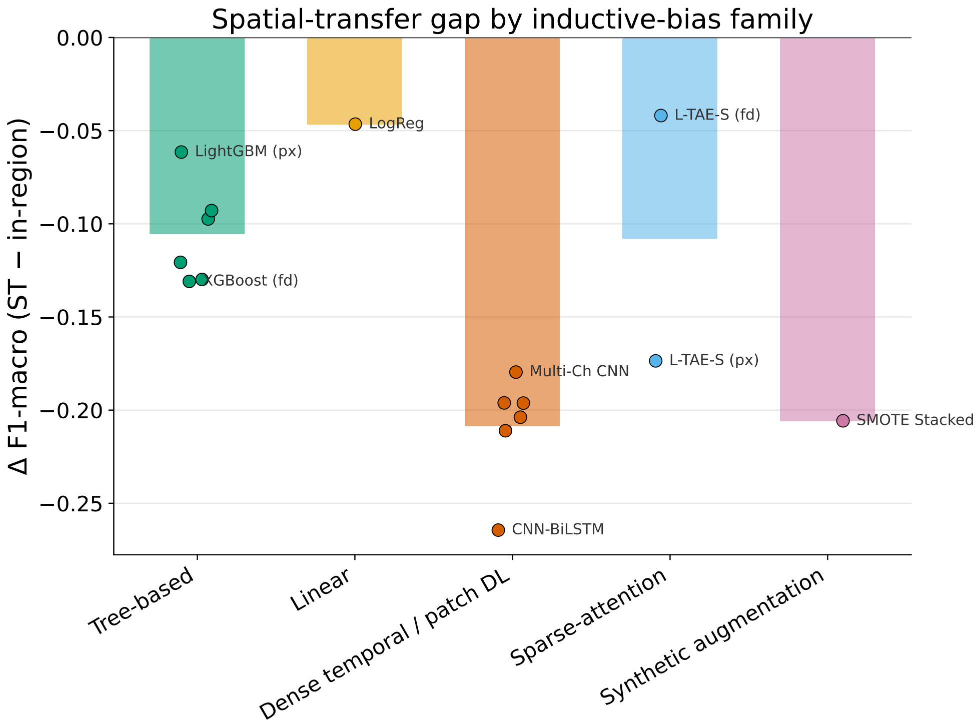

Fig. 15: Spatial-transfer gap by inductive-bias family.

Δ macro-F1 (spatial transfer minus in-region) grouped by inductive-bias family. Sparse/tree-like and linear models transfer (small $|\Delta|$); dense temporal/patch deep nets and synthetic augmentation overfit the training region. L-TAE-S, which adds sparse feature selection to a dense attention backbone, lies between L-TAE and the trees at the pixel level ($\Delta = -0.17$) and reaches the smallest gap in the study when field-aggregated ($\Delta = -0.04$); its family mean (-0.11) therefore tracks the tree-based models. Bars show family means; points are individual models.

available even when the architecture is fixed. A model that already has the first lever gains nothing, and may lose observation volume, from the second; this is exactly why TabNet prefers pixels. But the two levers also point to a novel solution: L-TAE-S adds a sparse feature gate to a temporal-attention backbone, then benefits further from field aggregation, ending with the smallest transfer gap in the benchmark. The fragile cluster (CNN-BiLSTM, 3D CNN, SMOTE stacking) combines high capacity, dense temporal representation, and (for SMOTE) synthetic interpolation, with neither lever engaged. Appendix D consolidates the five controlled comparisons behind this account and confirms that the remaining candidate explanations, ensemble size and the imbalance-handling objective, move holdout accuracy only at the margin and leave the robust–fragile ordering intact.

Practitioners can read the two levers as a flowchart. If you have the freedom to choose or modify the model, build feature selection into it (lever 1). If you are constrained to an off-the-shelf dense architecture, aggregate predictions to the field level instead (lever 2). Combining both is not free, because aggregation strips the per-pixel observations that a sparse model relies on. The broader point is that transfer-robustness is not the absence of expressivity; it is a model property that can be designed in, and the next two decades of remote-sensing deep learning should aim to do so deliberately rather than chase in-region scores and hope they generalize.

Automated feature extraction and computational cost.

The classical pipelines paired with `xr_fresh` were not only robust but cheap: the `xr_fresh` Voting and Stacking ensembles transfer as well as the best deep models (ST $\mathrm{F1}_m$ 0.55–0.56) while training in minutes, versus hours for the temporal deep nets (Table 3). To our knowledge this is the first evaluation of `xr_fresh` in a crop-classification context, and the result is encouraging: automated time-series feature extraction yields a representation that, fed to a tree ensemble, both matches deep models on transfer and inherits the trees' robustness, a favourable combination for operational, resource-constrained deployment.

Patch-level spatial modeling.

Patch-level convolutional models did not exceed pixel-level models in-region and transferred poorly: the 3D CNN reached $\mathrm{F1}_m = 0.50$ ($\Delta = -0.20$) and the Multi-Channel 2D CNN only $\mathrm{F1}_m = 0.45$ ($\Delta = -0.18$), placing both in the lower third of the spatial-transfer ranking. This is partly structural. At Sentinel-2's 10 m resolution the native 100 m patches are only $\sim$10$\times$10 pixels, so the local spatial context they can encode is limited, and resampling them to the network's 128×128 input adds interpolated rather than genuine detail. Narrow or small fields yield few fully-contained interior patches, further limiting the spatial signal available. Combined with a dense convolutional representation that offers no transfer-protective sparsity, these factors leave the patch models with little to exploit and among the weakest spatial-transfer performance of any paradigm.

Persistent difficulty with small-grain grazing and barley.

Small-grain grazing, barley, and wheat are the hardest classes for every model we examined (Figure 14), and they account for most of the accuracy lost when moving from the training region to the holdout. Small-grain grazing has both the fewest distinctive observations and a seasonal spectral signature that closely resembles the other cereals, particularly early in the season; the wheat–barley confusion is similarly persistent and consistent with their spectral overlap (Figure 4). That even the robust models struggle here, and that CNN-BiLSTM loses accuracy on *every* class (not only the rare ones), indicates the limitation is partly intrinsic similarity between these crops and partly the spatial-transfer problem itself, not class imbalance alone. Targeted label collection for these classes, and richer temporal sampling, are the most promising remedies.

Limitations.

Several limitations constrain these results. First, the study uses a single growing season (2017) and a single holdout tile that is spatially *adjacent* to the training tiles and lies in the same agroecological zone and growing calendar. This is therefore a relatively local spatial shift, and the gaps we report most likely *understate* the degradation that fully cross-regional or cross-year transfer would produce, so our numbers are best read as a conservative lower bound. That even this mild shift reorders the models is precisely the cautionary point, but multi-region and multi-year holdouts are needed to confirm that the inductive-bias ordering is stable across larger distribution shifts; we accordingly frame our finding as a method-level conclusion about transfer rather than a claim about absolute achievable accuracy on this small benchmark. Second, the labels derive from a public Radiant Earth MLHub dataset rather than our own field campaign; this aids reproducibility but means we inherit its digitizing and survey conventions. Third, we use Sentinel-2 optical data only. The exclusion of SAR is a deliberate scoping choice (to isolate the optical time-series signal and maximize applicability where only optical data exist), but it limits the achievable ceiling, particularly for optically similar cereals; SAR fusion is a clear next step (below).

6 Conclusion

The headline result of this study is methodological: in-region validation, the protocol the crop-classification literature reports almost universally, misranks twenty-three Sentinel-2 classifiers when they are applied to a spatially disjoint holdout. Dense temporal networks that lead under conventional evaluation become near-worst under transfer, while parsimonious models such as tree ensembles, logistic regression, and a TabNet ensemble retain rank and accuracy. The reordering is large enough that "best model" is, in this setting, an artifact of the validation protocol.

The mechanism behind the reordering is, we argue, the same property that made tree ensembles the remote-sensing default for two decades: a sparse, axis-aligned feature-selection bias that commits to a few robust signals and ignores the rest. TabNet inherits this bias through sparsemax masks; a Transformer encoder with far greater attention capacity does not, and transfers no better than the lightweight L-TAE, pointing at sparsity, not capacity, as the lever. Grafting that lever onto an attention model (our L-TAE-S) ties TabNet for the best transfer performance, and at the field level posts the smallest in-region-to-holdout gap of any model in the benchmark ($\Delta = -0.04$). Field-level aggregation is a second, partly substitutable lever (variance reduction through pixel averaging) that composes with sparsity in L-TAE-S but hurts TabNet, whose sparse gate exploits the per-pixel observation volume that aggregation removes.

These findings translate into three concrete recommendations for the field. First, evaluate on a spatially disjoint holdout before declaring one architecture better than another; in-region cross-validation, however carefully segmented, will not catch the reordering reported here. Second, when the goal is mapping unseen terrain, prefer models with a built-in feature-selection bias over expressive dense ones; the transfer cost of expressivity is large and predictable. Third, if dense temporal models are required, aggregate their predictions to the field level; this recovers most of the lost transfer accuracy at no architectural cost. Across all models, small-grain grazing and the wheat–barley confusion remain the dominant residual errors, pointing to richer temporal sampling and targeted collection for these classes as the most direct route to further gains. The most productive future directions are multi-region and multi-year holdouts to test the stability of the inductive-bias ordering, SAR–optical fusion to raise the achievable ceiling for spectrally similar cereals, and domain-adaptation methods that explicitly optimize for transfer. Higher-cadence optical platforms such as PlanetScope could further mitigate the cloud-cover and temporal-resolution limits encountered here.

More broadly, the conditions that make the Western Cape difficult, such as scarce and uneven crop labels, heterogeneous smallholder and dryland management, irregular planting calendars, and persistent cloud cover, are the conditions under which agricultural remote sensing must operate across much of the Global South, where crop maps inform food-security and climate-resilience decisions for billions of people. Studying methods in this setting is not a niche stress test but a more realistic account of how remote sensing performs when it leaves the conditions in which it was developed. Evaluating models the way they are actually deployed (applied to regions where no labels were collected) and favouring parsimonious, data-efficient classifiers that hold up under that transfer is, we argue, what will make remote sensing deliver in precisely the settings where it matters most.

Conflict of Interest

The authors declare that they have no competing interests.

Data Availability

This study relied on multiple sources of data. Ground-truth crop type labels were obtained from Radiant Earth's MLHub platform [2], while multi-temporal Sentinel-2 imagery was extracted using the Google Earth Engine API. Since the raw data was processed and combined in various ways specific to this study, the resulting dataset used for model training and evaluation is not hosted publicly but can be shared by the authors upon reasonable request for academic or research purposes.

Code Availability

All source code used for data processing, modeling, and evaluation is publicly available [4].

A Baseline classical model definitions

For completeness we reproduce the standard formulations of the four baseline classifiers described in Section 3.1. For a feature vector $\mathbf{x}_i$ and class $c \in \{1, \ldots, C\}$:
Logistic Regression [25] (multinomial / softmax):

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{x}_i + b_c)}{\sum_{c'=1}^{C} \exp(\mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'})},$$

with per-class weight vectors $\mathbf{w}_c$ and biases b_c.

Random Forest [26]: majority vote over K decision trees,

$$\hat{y}_i = \operatorname{mode}\left\{T_1(\mathbf{x}_i), \ldots, T_K(\mathbf{x}_i)\right\}.$$

LightGBM [27]: additive gradient boosting with learning rate η and tree f_t,

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \cdot f_t(\mathbf{x}_i).$$

XGBoost [28, 29]: regularized boosting objective,

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \qquad \Omega(f_k) = \gamma T_k + \tfrac{1}{2}\lambda \sum_j w_j^2,$$

where l is the per-sample loss, T_k the number of leaves in tree f_k, w_j the leaf weights, and γ, λ regularization coefficients.

B Performance metric definitions

Let N be the number of samples, C the number of classes, and n_c the support of class c. Cohen's Kappa is

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_o = \frac{\#\text{correct}}{N}, \quad p_e = \sum_{c=1}^{C} \frac{n_{c,\text{true}}}{N} \cdot \frac{n_{c,\text{pred}}}{N}.$$

Macro-averaged F1 (our primary metric) and weighted F1 are

$$\mathrm{F1}_m = \frac{1}{C}\sum_{c=1}^{C} \mathrm{F1}_c, \qquad \mathrm{F1}_{\text{weighted}} = \sum_{c=1}^{C} \frac{n_c}{N}\,\mathrm{F1}_c, \qquad \mathrm{F1}_c = \frac{2\,\mathrm{Prec}_c\,\mathrm{Rec}_c}{\mathrm{Prec}_c + \mathrm{Rec}_c}.$$

Cross-entropy (Xent). This is the official field-level scoring metric of the AI4FoodSecurity (South Africa) challenge [3], in which our field-label dataset was used: a cross-entropy with a binary outcome for each crop. Because challenge submissions consist of a single predicted crop ID per field (a hard label, not a probability vector), the metric is evaluated on the one-hot encoding of the predicted label, clipped to $[\epsilon, 1-\epsilon]$ with $\epsilon = 10^{-7}$, against the one-hot true label:

$$\mathrm{Xent} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\,\ln\bigl(\hat{q}_{i,c}\bigr), \qquad \hat{q}_{i,c} = \operatorname{clip}\bigl(\mathbb{1}[\hat{y}_i = c],\, \epsilon,\, 1-\epsilon\bigr),$$

where $y_{i,c}$ is the one-hot true label and $\hat{y}_i$ the predicted class (lower is better). We adopt it so that out-of-sample performance is also expressed on the source challenge's standardized, operationally-defined metric, alongside the balanced-accuracy measures above.

C Training cost

Table 6 reports the wall-clock training time for a single training run of each model, measured on one workstation (NVIDIA RTX 3080 Ti GPU; deep networks and the GPU-accelerated gradient-boosted trees use the GPU, while the scikit-learn Random Forest and Logistic Regression run on CPU). Times exclude feature extraction (`xr_fresh`) and data loading, and are intended to convey order-of-magnitude differences in computational cost across paradigms rather than exact benchmarks. For the multi-seed ensembles the figure is the mean time for one seed; the full ensemble costs roughly five times as much.

The pattern reinforces the computational-cost argument made in Section 4 and Section 5: classical tree and linear learners on the compact `xr_fresh` representation train in seconds to a few minutes at the field level, whereas the dense temporal and patch deep networks require tens of minutes to hours per seed even on a GPU. The robust models that transfer best (the `xr_fresh` tree ensembles and the sparse-attention L-TAE-S at the field level) are also among the cheapest to train, while the most expensive models (TabTransformer, the patch CNNs, the pixel-level Transformer) do not lead the holdout ranking.

Table 6: Training time for a single run (one seed) of each model.

Model	Level	Features	Time / run
Classical, field-level			
XGBoost	field	`xr_fresh`	12 s
LightGBM	field	`xr_fresh`	31 s
Voting	field	`xr_fresh`	109 s
Stacking	field	`xr_fresh`	109 s
SMOTE Stacked	field	`xr_fresh`	99 s
Classical, pixel-level			
Base XGBoost	pixel	`xr_fresh`	31 s
Base LightGBM	pixel	`xr_fresh`	31 s
Logistic Regression	pixel	`xr_fresh`	8.0 min
Random Forest	pixel	`xr_fresh`	73 min
Deep, field-level (raw sequences)			
TabNet (`xr_fresh`)	field	`xr_fresh`	34 s
TabNet (temporal)	field	raw	39 s
CNN-BiLSTM	field	raw	53 s
L-TAE	field	raw	61 s
TempCNN	field	raw	64 s
L-TAE-S	field	raw	18 s†
TabTransformer	field	`xr_fresh`	7.4 h
Deep, pixel-level (raw sequences)			
CNN-BiLSTM	pixel	raw	10 min†
L-TAE-S	pixel	raw	24 min†
TempCNN	pixel	raw	29 min
L-TAE	pixel	raw	38 min
TabNet	pixel	raw	61 min†‡
Transformer	pixel	raw	97 min
Deep, patch-level (raw sequences)			
3D CNN	patch	raw	22 min†
Multi-Channel CNN	patch	raw	38 min†

The gradient-boosted trees (XGBoost, LightGBM) are timed as a single best-configuration fit, excluding the one-time 150-trial Optuna hyperparameter search (roughly 1.8 h and 3.4 h respectively), so that they are comparable to the single-fit pixel-level trees. †Mean time per seed for a 5-seed ensemble. ‡Measured on the 75% training fraction, the best-transferring TabNet pixel configuration used in Table 3.

D Ablation studies

A natural question is whether the spatial-transfer results of Section 4 are driven by architecture, by ensembling, by prediction aggregation, or by imbalance handling. We answer it with five controlled comparisons, three already embedded in the main text and two added here, each varying one factor while holding the others fixed. Read together they attribute the holdout ranking primarily to two architectural and representational levers — built-in sparse feature selection and field-level aggregation — and show that ensemble size and the imbalance-handling objective are second-order: they shift absolute scores but never reorder the robust and fragile model groups.

(i) Field-level aggregation.

Running the same architecture on per-pixel versus field-averaged inputs isolates aggregation as a variance-reduction lever (Section 4.2, Table 4). It recovers most of the transfer gap for dense temporal networks — L-TAE's gap shrinks from $\Delta = -0.20$ (pixel) to -0.07 (field) and L-TAE-S's from -0.17 to -0.04, the smallest of any model — while being roughly neutral for the already-sparse tree learners and actively *harmful* for TabNet, whose $F1_m$ falls from 0.60 to 0.43 because field training starves its sparse gate of the per-pixel observation volume it exploits.

(ii) Sparse feature selection versus attention capacity.

A plain Transformer encoder with far greater attention capacity than L-TAE transfers no better than it (both ST $F1_m = 0.58$, $\Delta = -0.20$ at the pixel level; Table 3), whereas grafting a TabNet-style sparse channel gate onto the same lightweight backbone (L-TAE-S) lifts transfer to $F1_m = 0.60$ and, when field-aggregated, to the smallest gap in the benchmark. Capacity is not the lever; sparse feature selection is.

(iii) Resampling.

The *Stacking* and *SMOTE Stacked* ensembles share an identical architecture and differ only in SMOTETomek resampling, isolating class rebalancing. Resampling *widens* the transfer gap (from $\Delta = -0.11$ to -0.21) and lowers holdout $F1_m$ (from 0.53 to 0.49; Table 3): synthetic minority interpolation aids in-region fit but hurts spatial transfer.

(iv) Ensemble size.

Table 7 sweeps the seed-ensemble size (1/3/5) for three field-level models by re-inference of the saved per-seed checkpoints. Ensembling gives a small, monotone lift (L-TAE-S rises from $F1_m = 0.56$ at one seed to 0.60 at five) but does not change the ordering: a *single*-seed L-TAE-S (mean $F1_m = 0.56$) already exceeds the five-seed CNN-BiLSTM and TabNet field ensembles (0.51 and 0.43). The robustness ranking is therefore not an artifact of ensembling.

(v) Imbalance-handling objective and resampling.

Table 8 retrains each field-level model under five imbalance-handling settings: weighted focal loss ($\gamma = 2$, the main-text default), class-weighted cross-entropy ($\gamma = 0$), plain unweighted cross-entropy, and the focal and plain losses each combined with class-balanced oversampling. The robust L-TAE-S is almost invariant to these choices (ST $F1_m$ in 0.59–0.60 across all five), whereas the fragile CNN-BiLSTM is sensitive to them ($F1_m$ ranging 0.41–0.56): imbalance handling matters most where the architecture is weakest. Crucially, even CNN-BiLSTM's best setting (0.56) remains below L-TAE-S's worst (0.59), so the objective never reorders the two. TabNet is shown at the field level, where retraining is cheap; its flagship pixel-level ensemble — the best transferer overall at $F1_m = 0.60$ — is reported in Table 3.

Table 7: Ablation (iv): spatial-transfer holdout performance versus seed-ensemble size, by re-inference of the saved per-seed checkpoints. The single-seed row reports the mean±standard deviation across the five individual seeds; the 3-seed row uses seeds $\{42, 101, 202\}$ and the 5-seed row the full set. The 5-seed rows reproduce the main-text configuration.

Model	**Ensemble**	$\mathbf{F1}_m$	κ
TabNet (field)	1 seed	0.31 ± 0.02	0.18 ± 0.03
	3 seeds	0.39	0.29
	5 seeds	0.43	0.34
L-TAE-S (field)	1 seed	0.56 ± 0.02	0.47 ± 0.02
	3 seeds	0.58	0.50
	5 seeds	0.60	0.52
CNN-BiLSTM (field)	1 seed	0.48 ± 0.02	0.31 ± 0.02
	3 seeds	0.52	0.34
	5 seeds	0.51	0.33

Table 8: Ablation (v): effect of the imbalance-handling objective and resampling on spatial-transfer holdout performance (field-level retrains, 5-seed ensembles). The baseline (weighted focal, γ=2, no resampling) is the main-text configuration. "Resample" denotes class-balanced oversampling of the training set: a `WeightedRandomSampler` with inverse-frequency weights for the neural models, and a one-shot balanced oversample for TabNet, whose library exposes no equivalent sampler hook.

Model	**Loss**	**Resample**	$\mathbf{F1}_m$	κ
TabNet (field)	Weighted focal (γ=2)	no	0.43	0.34
	Weighted CE (γ=0)	no	0.42	0.33
	Plain CE	no	0.35	0.25
	Weighted focal (γ=2)	yes	0.40	0.23
	Plain CE	yes	0.46	0.37
L-TAE-S (field)	Weighted focal (γ=2)	no	0.60	0.52
	Weighted CE (γ=0)	no	0.60	0.50
	Plain CE	no	0.59	0.50
	Weighted focal (γ=2)	yes	0.60	0.49
	Plain CE	yes	0.60	0.50
CNN-BiLSTM (field)	Weighted focal (γ=2)	no	0.51	0.34
	Weighted CE (γ=0)	no	0.49	0.34
	Plain CE	no	0.41	0.30
	Weighted focal (γ=2)	yes	0.56	0.39
	Plain CE	yes	0.50	0.34

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